

Memory, Roughness, and Information Persistence in Financial Markets: A Structural Approach to Volatility Forecasting.

Akash Deep,^{1,*} Nicholas Appiah² and Svetlozar T. Rachev³

¹Department of Mathematics and Statistics, Texas Tech University, Lubbock, 79409, Texas, USA

²Department of Mathematics and Statistics, Texas Tech University, Lubbock, 79409, Texas, USA

³Department of Mathematics and Statistics, Texas Tech University, Lubbock, 79409, Texas, USA

*Corresponding author: akash.deep@ttu.edu

Abstract

This paper studies the joint role of long-memory dynamics, rough-volatility behavior, and persistence-based forecasting features in equity volatility modeling. Using 115 S&P 500 constituents from November 2001 to April 2026, we combine semiparametric long-memory estimation, rolling rough-volatility diagnostics, and structured forecasting regressions. Volatility proxies exhibit substantial persistence and local roughness, with cross-sectional mean GPH and local-Whittle memory estimates of 0.226 and 0.440, respectively. Rolling persistence rises during the global financial crisis and COVID and is positively associated with the VIX. Out-of-sample forecasts show that cross-sectional persistence aggregates, sectoral persistence measures, and persistence-by-stress interactions improve HAR and HAR-X benchmarks, especially at longer horizons and during stress regimes. Volatility-managed portfolio results further indicate that these gains are economically meaningful and robust across key model variations and evaluation settings. Persistence measures therefore serve as reduced-form indicators of the duration and propagation of financial uncertainty, without claiming structural identification of the underlying mechanisms.

Keywords Long memory, rough volatility, volatility forecasting, financial econometrics, realized volatility, persistence dynamics

JEL Classification: C22, C53, G12, G17

Introduction

Volatility forecasting occupies a central role in financial econometrics because volatility enters risk management, derivative pricing, portfolio allocation, margin determination, stress testing, and macro-financial monitoring. A large empirical literature has documented that volatility measures exhibit substantially greater persistence than raw asset returns. Classical long-memory models formalized this observation by extending conventional weak-dependence frameworks to allow autocorrelations that decay at hyperbolic rather than exponential rates (Granger and Joyeux, 1980; Hosking, 1981). Subsequent work incorporated fractional dependence into conditional-volatility specifications through FIGARCH-type models (Baillie et al., 1996), while the realized-volatility literature improved the empirical measurement of latent volatility dynamics using high-frequency data (Andersen et al., 2003; Barndorff-Nielsen and Shephard, 2002).

Despite extensive empirical evidence for persistence in volatility proxies, the interpretation of persistence remains controversial. One branch of the literature views long-memory behavior as reflecting structural characteristics of financial markets, including heterogeneous trading horizons, slow information diffusion, and

institutional frictions (Müller et al., 1997). Another branch argues that apparent long-range dependence may arise from structural breaks, stochastic regime changes, or nonstationary volatility dynamics rather than genuine fractional dependence (Mikosch and Stărică, 2004). As a result, the economic interpretation of persistence measures remains unsettled. The central issue is therefore not simply whether a fractional parameter can be estimated, but whether persistence measures behave coherently across market states and contain useful information for forecasting and risk management.

This paper examines volatility persistence from a reduced-form perspective. Rather than interpreting the memory parameter as a purely technical object associated with fractional differencing, we investigate whether rolling persistence measures behave empirically like indicators of the temporal propagation of uncertainty in financial markets. More specifically, we study whether persistence measures rise during sustained stress episodes, whether they co-move with broad market uncertainty indicators, and whether they improve volatility forecasts beyond standard benchmark models. The analysis does not attempt to identify a unique structural mechanism generating persistence. Instead, persistence is treated as an empirical summary statistic whose dynamics may reflect several overlapping mechanisms, including heterogeneous investor horizons,

liquidity conditions, delayed information processing, regime shifts, and market-wide stress propagation.

The paper also considers the relationship between long-memory behavior and rough volatility. Recent research has shown that volatility paths often display substantial local irregularity, with estimated Hurst exponents significantly below the Brownian benchmark of $\frac{1}{2}$ (Gatheral et al., 2018; Bennedsen et al., 2021). Although roughness and long memory are sometimes presented as competing explanations of volatility dynamics, they describe different temporal characteristics. Long memory concerns the persistence of dependence over long horizons, whereas roughness concerns local path irregularity at very short horizons. The empirical analysis in this paper suggests that both characteristics may coexist within the same volatility process.

A further issue concerns the role of machine learning in volatility forecasting. Flexible forecasting algorithms can capture nonlinear interactions and high-dimensional predictor structures that may be difficult to represent using conventional parametric models. However, purely data-driven forecasting systems often lack economic interpretability and may produce unstable results outside the estimation sample (Gu et al., 2020). The present paper adopts an intermediate approach: econometric methods are used to construct interpretable persistence-related features, while forecasting systems incorporate these features alongside conventional volatility predictors and market-state variables.

The empirical analysis uses a panel of 115 S&P 500 stocks over the period November 2001 through April 2026. We estimate rolling long-memory measures using both the Geweke–Porter–Hudak and local-Whittle estimators and estimate local roughness using rolling Hurst exponents. These variables are incorporated into a layered forecasting framework that compares HAR, HAR-X, persistence-augmented regressions, shrinkage estimators, and tree-based machine-learning methods.

The empirical results may be summarized as follows. First, volatility proxies exhibit substantial long-memory behavior and strong local roughness throughout the panel. Second, rolling persistence measures rise markedly during periods of market stress and display positive association with the VIX. Third, persistence-related forecasting features provide statistically significant incremental improvements beyond HAR and HAR-X benchmarks, particularly during stress regimes and at longer horizons. Fourth, the gains arise primarily from cross-sectional and interaction effects rather than from own-stock persistence alone. Finally, linear persistence-augmented systems outperform tree-based machine-learning methods, suggesting that economically structured feature design matters more than estimator complexity.

The remainder of the paper proceeds as follows. Section 2 reviews the literature. Section 3 discusses the theoretical foundations of persistence. Section 4 presents the econometric framework. Section 5 develops the reduced-form interpretation. Section 6 introduces the forecasting framework. Section 7 describes the data and empirical design. Section 8 presents the results. Section 9 discusses the economic implications. Section 10 concludes.

Literature Review: Persistence, Roughness, and Volatility

The literature on volatility persistence originates with foundational work on fractional integration and ARFIMA processes, which extended classical time-series analysis by allowing autocorrelation structures with hyperbolically decaying dependence rather than conventional exponential decay (Granger and Joyeux, 1980; Hosking, 1981). These developments became especially important in financial econometrics because volatility proxies such as squared returns, absolute returns, and realized variance measures exhibit substantially stronger persistence than raw asset returns themselves (Ding et al., 1993). The resulting evidence motivated the development of models capable of capturing long-range dependence in financial volatility while remaining consistent with broader conditional heteroskedasticity frameworks.

One important contribution in this direction was the FIGARCH model introduced by Baillie et al. (1996), which incorporated fractional integration directly into conditional variance dynamics. FIGARCH generalized conventional GARCH specifications by allowing shocks to decay at hyperbolic rates, thereby accommodating the empirically observed persistence of volatility. The model rapidly became a standard benchmark in long-memory volatility research because it provided a tractable parametric representation of persistent conditional heteroskedasticity. Subsequent extensions introduced asymmetric responses, multivariate formulations, and alternative fractional structures, further expanding the long-memory volatility literature.

At approximately the same time, the realized-volatility literature substantially improved the empirical measurement of latent volatility processes. Andersen et al. (2003) and Barndorff-Nielsen and Shephard (2002) demonstrated that high-frequency financial data could be used to construct more accurate *ex post* measures of integrated volatility. These developments strengthened empirical evidence for volatility persistence because realized-volatility measures contain less measurement noise than traditional squared-return proxies. The resulting data environment allowed researchers to study persistence properties with considerably greater precision.

Another important development was the HAR model proposed by Corsi (2009). Rather than imposing exact fractional integration, the HAR framework approximates long-memory behavior using a parsimonious multi-horizon structure involving daily, weekly, and monthly realized-volatility components. The HAR model proved empirically successful because it combines forecasting accuracy with economic interpretability. In particular, it is consistent with the idea that financial markets contain heterogeneous agents operating at different investment horizons (Müller et al., 1997). The HAR framework remains one of the most widely used benchmarks in volatility forecasting.

Despite substantial empirical support for persistence, the interpretation of persistence remains controversial. One branch of the literature views long memory as reflecting genuine structural characteristics of financial markets, including heterogeneous trading frequencies, delayed information diffusion, and institutional frictions. Another branch argues that apparent long-range dependence may arise from structural breaks, regime changes, stochastic volatility shifts, or nonlinear dynamics rather than from genuine fractional integration (Mikosch and Stărică, 2004). This critique implies that estimated persistence parameters should be interpreted cautiously

because they may reflect evolving market conditions rather than stable stochastic mechanisms.

The debate surrounding long memory became even more important following the emergence of the rough-volatility literature. Gatheral et al. (2018) showed that volatility paths frequently exhibit substantial local irregularity, with estimated Hurst exponents significantly below the Brownian-motion benchmark of $H = \frac{1}{2}$. Subsequent work clarified that roughness and long memory are not necessarily competing concepts. Bennedsen et al. (2021) argued that short-run roughness and long-horizon persistence may represent distinct temporal features of volatility processes. The empirical analysis in this paper adopts precisely this perspective by estimating both rolling persistence measures and rolling Hurst exponents within the same forecasting framework.

In parallel with these econometric developments, machine learning has become increasingly important in volatility forecasting and empirical asset pricing. Flexible learning algorithms are capable of capturing nonlinear interactions, threshold effects, and high-dimensional predictor structures (Breiman, 2001; Friedman, 2001; Gu et al., 2020; Hastie et al., 2009). However, purely data-driven approaches may produce unstable results outside the estimation sample and frequently lack economic interpretability.

This tension between statistical flexibility and economic interpretation has become a central issue in modern empirical finance. Recent research increasingly emphasizes hybrid approaches in which economically motivated features are combined with flexible forecasting architectures. In this setting, the quality and interpretability of the feature space may be more important than the complexity of the forecasting algorithm itself.

Taken together, the literature reveals several unresolved issues. First, there remains substantial disagreement regarding the economic interpretation of volatility persistence and its relationship to regime dependence, structural breaks, and heterogeneous trading behavior. Second, the relationship between long-memory dynamics and rough-volatility behavior is still not fully understood. Third, relatively few studies examine whether persistence measures themselves contain incremental forecasting information beyond conventional volatility predictors and market-state variables. The present paper contributes by examining persistence measures from a reduced-form empirical perspective, studying whether rolling persistence measures behave coherently across market regimes and provide economically meaningful forecasting information.

Theoretical Foundations of Long Memory and Information Persistence

The concept of long memory emerged from attempts to describe stochastic processes whose dependence structures cannot be adequately represented by conventional short-memory models. Long-memory processes retain measurable dependence over substantially longer horizons, typically characterized through a slowly decaying autocovariance structure in which dependence declines at a hyperbolic rather than exponential rate (Granger and Joyeux, 1980; Hosking, 1981).

Formally, a covariance-stationary long-memory process may be characterized by an autocovariance function satisfying

$$\gamma(k) \sim c_\gamma k^{2d-1} \quad \text{as } k \rightarrow \infty, \quad (1)$$

where $d \in (0, \frac{1}{2})$ denotes the memory parameter.

Because $2d - 1 > -1$, the autocovariances are not absolutely summable, implying persistent dependence across long horizons. Equivalent characterizations arise in the frequency domain,

$$f(\lambda) \sim c_f \lambda^{-2d}, \quad \lambda \rightarrow 0, \quad (2)$$

providing the foundation for semiparametric estimators of long memory, including the Geweke–Porter–Hudak and local-Whittle procedures employed later in the paper (Robinson, 1995).

The fractional-difference operator $(1 - L)^d$ provides the bridge between these probabilistic properties and empirically estimable time-series models. Fractionally integrated ARFIMA systems express current observations as weighted combinations of past innovations, where the weights decay gradually rather than disappearing rapidly. Relative to standard ARMA structures, fractional integration allows shocks to persist over much longer horizons while preserving covariance stationarity when $d < \frac{1}{2}$. In financial econometrics, this framework proved especially useful because volatility measures routinely display persistence patterns that are inconsistent with short-memory dynamics.

Although long-memory models originated as statistical descriptions of temporal dependence, their interpretation in financial applications naturally raises broader economic questions. When persistence is observed in volatility proxies rather than in returns themselves, the memory parameter may be viewed as summarizing the duration over which uncertainty shocks continue to influence market behavior. Importantly, this interpretation should not be understood as structural identification of a unique mechanism. Rather, persistence measures are best viewed as reduced-form summaries reflecting several overlapping economic forces that jointly influence the temporal propagation of volatility.

One important explanation for persistent volatility dynamics arises from heterogeneous market participation. Financial markets contain agents operating at different horizons, using different information sets, and responding at different speeds to new information. High-frequency traders may react within milliseconds to order-flow imbalances and liquidity conditions, whereas institutional investors may rebalance portfolios over days, weeks, or months. Pension funds, insurance companies, and macro-oriented asset managers frequently adjust positions gradually in response to evolving risk conditions rather than immediately after news arrivals. The aggregation of these heterogeneous responses can generate long-horizon persistence even when individual agents themselves do not follow fractional stochastic laws (Müller et al., 1997; Corsi, 2009).

Another important channel involves delayed information processing and gradual uncertainty resolution. Macroeconomic announcements, monetary-policy surprises, geopolitical shocks, or systemic financial disturbances may become publicly observable immediately, yet their implications for risk, liquidity, hedging demand, and portfolio reallocation may unfold only gradually. Under such conditions, volatility persistence reflects not only the arrival of information but also the prolonged process through which financial markets absorb, interpret, and respond to uncertainty. In this setting, persistence measures may summarize the speed at which uncertainty dissipates throughout the financial system.

Market microstructure mechanisms provide an additional source of persistence. Order splitting, inventory management by dealers, intermittent liquidity provision, and interactions between informed

and uninformed traders can generate volatility clustering at short horizons. Repeated microstructure frictions may then accumulate into broader dependence patterns observable over longer periods. Under this interpretation, long-memory behavior represents the macro-level consequence of repeated local frictions and trading adjustments rather than a purely abstract time-series property.

Behavioral mechanisms may also contribute to persistent volatility dynamics. Financial markets do not always process information instantaneously or efficiently. Underreaction, overreaction, limited attention, ambiguity aversion, and delayed belief updating can all produce gradual adjustment processes in which shocks remain economically relevant for extended periods. Although the classical long-memory literature was not originally formulated within behavioral-finance frameworks, the empirical phenomenon of slow volatility adjustment is consistent with several behavioral explanations of market dynamics.

These considerations suggest that persistence measures may serve as reduced-form indicators of the temporal organization of uncertainty in financial markets. A stylized representation may be written as

$$I_t = \sum_{k=0}^{\infty} w_k \varepsilon_{t-k}, \quad w_k \sim c k^{d-1}, \quad (3)$$

where I_t denotes an uncertainty-related process influenced by past informational shocks ε_{t-k} , and d governs the speed at which the influence of past shocks declines over time.

It is important, however, to distinguish carefully between statistical persistence and economic interpretation. The presence of estimated long memory does not necessarily imply that markets literally follow stationary fractional stochastic laws. Structural breaks, regime changes, stochastic volatility shifts, and evolving institutional environments may generate empirical behavior resembling long-range dependence. Consequently, persistence measures should not be interpreted mechanically as identifying a single structural mechanism. Instead, they should be viewed as empirical indicators whose behavior may reflect several interacting economic channels simultaneously.

The interpretation developed in this paper therefore remains intentionally reduced-form. The central empirical question is not whether a unique structural mechanism can be identified, but rather whether persistence measures behave coherently across market states and whether they contain useful forecasting information beyond conventional volatility predictors. If persistence measures rise systematically during sustained stress episodes, co-move with broad uncertainty indicators, and improve volatility forecasts during crisis periods, then they may provide economically informative summaries of market conditions even without full structural identification.

This perspective also accommodates the rough-volatility literature. Roughness and long memory describe different aspects of temporal dependence rather than mutually exclusive mechanisms. Long memory concerns low-frequency dependence and the persistence of shocks across longer horizons. Roughness concerns local path irregularity at very short horizons. A rough-volatility process satisfies

$$E|\log \sigma_{t+\Delta}^2 - \log \sigma_t^2|^q \propto \Delta^{qH}, \quad (4)$$

where $H < \frac{1}{2}$ implies rougher behavior than standard Brownian motion (Gatheral et al., 2018). A volatility process may therefore be simultaneously rough at short horizons and persistent at longer horizons. Financial markets may react sharply and irregularly

immediately after news arrivals while still exhibiting prolonged uncertainty propagation over subsequent periods.

The coexistence of roughness and persistence suggests that volatility dynamics should be analyzed across multiple temporal scales rather than through a single dependence measure. High-frequency volatility fluctuations may primarily reflect microstructure effects and rapid information arrivals, whereas medium- and long-horizon persistence may reflect broader institutional adjustment processes, funding conditions, and macro-financial uncertainty propagation. The empirical framework developed later in the paper incorporates both dimensions simultaneously through rolling estimates of long-memory parameters and rolling Hurst exponents.

Finally, the interpretation of persistence developed here has practical implications for volatility forecasting and risk management. If persistence measures summarize the expected duration of uncertainty conditions, then they may contain forecasting information that differs conceptually from conventional volatility-level indicators such as the VIX. Two periods with similar contemporaneous volatility levels may nevertheless differ substantially in the expected duration of uncertainty propagation. Persistence measures may therefore provide complementary information regarding how long elevated volatility conditions are likely to persist.

The empirical sections of the paper investigate this possibility directly by incorporating persistence-related features into structured forecasting systems and examining whether they improve forecasting performance across market regimes, sectors, and forecast horizons.

Econometric Modelling of Fractional and Rough Volatility

The econometric analysis of volatility persistence has evolved along two closely related but conceptually distinct directions. The first focuses on long-memory dynamics and emphasizes slow hyperbolic decay over medium and long horizons. The second focuses on rough volatility and emphasizes highly irregular local behavior at very short horizons. The long-memory literature developed primarily within discrete-time econometrics through fractional integration and ARFIMA-type models. Rough-volatility models, by contrast, emerged from continuous-time stochastic-volatility theory and high-frequency financial econometrics. Recent empirical evidence suggests that these two perspectives are not mutually exclusive. The empirical framework developed in this paper therefore incorporates both dimensions in a unified forecasting environment.

ARFIMA, GARCH, and FIGARCH

The ARFIMA framework provides a natural discrete-time representation of fractional dependence: for a covariance-stationary process $\{X_t\}$,

$$\phi(L)(1-L)^d X_t = \theta(L)\eta_t, \quad (5)$$

where $\phi(L)$ and $\theta(L)$ are finite-order lag polynomials, d denotes the fractional integration parameter, and η_t is a white-noise innovation process (Granger and Joyeux, 1980; Hosking, 1981). The operator $(1-L)^d$ generalizes conventional integer differencing to fractional orders and allows shocks to decay gradually over long horizons. When applied to volatility proxies such as squared returns, absolute returns, range-based volatility measures, or realized variance, ARFIMA models provide parsimonious representations of

long-range dependence in financial volatility. Unlike standard short-memory models, fractional integration allows dependence to persist over extended periods while preserving covariance stationarity when $d < \frac{1}{2}$.

The conditional heteroscedasticity literature approached volatility persistence from a different perspective. Beginning with the ARCH model of Engle (1982) and the GARCH extension of Bollerslev (1986), volatility dynamics were modeled directly through evolving conditional variance equations. Standard GARCH systems successfully capture volatility clustering but imply geometric decay in the effects of shocks. Empirical volatility series, however, often display substantially slower decay than conventional GARCH structures permit. The FIGARCH model of Baillie et al. (1996) addressed this limitation by embedding fractional integration directly within the conditional-variance recursion,

$$[1 - \beta(L)] h_t = \omega + [1 - \beta(L) - \phi(L)(1 - L)^d] \varepsilon_t^2, \quad (6)$$

where h_t denotes conditional variance and ε_t denotes return innovations. FIGARCH allows volatility shocks to decay hyperbolically rather than exponentially while preserving the conditional-heteroskedasticity structure.

The FIGARCH framework became particularly influential because it provided an explicit connection between long-memory econometrics and volatility modeling. Numerous subsequent extensions introduced asymmetric effects, multivariate specifications, nonlinear dynamics, and regime dependence. Nevertheless, the central empirical motivation remained unchanged: volatility shocks appear to dissipate much more slowly than conventional short-memory volatility models predict.

An important practical issue concerns interpretation. Although ARFIMA and FIGARCH systems estimate similar persistence behavior, they represent different modeling philosophies. ARFIMA-type systems emphasize the dependence structure of the observed process itself, whereas GARCH-type systems emphasize the evolution of latent conditional variance. In practice, both approaches frequently generate evidence consistent with persistent volatility dynamics.

Realized Volatility and HAR Models

The development of realized-volatility methodology substantially improved empirical analysis of volatility persistence. Traditional volatility proxies based on squared daily returns contain substantial measurement noise and therefore obscure underlying dependence structures. High-frequency financial data made it possible to construct more accurate ex post measures of integrated volatility through realized-variance estimators (Andersen et al., 2003; Barndorff-Nielsen and Shephard, 2002). Realized-volatility measures significantly strengthened empirical evidence for persistent volatility dynamics because they provide less noisy approximations to latent volatility processes. These measures also facilitated more precise estimation of long-memory behavior and improved volatility forecasting performance across a wide range of applications.

An especially important contribution was the heterogeneous autoregressive (HAR) model proposed by Corsi (2009). The HAR framework approximates long-memory behavior using a parsimonious multi-horizon structure rather than imposing exact fractional

integration,

$$\log RV_{t+1} = \beta_0 + \beta_d \log RV_t^{(d)} + \beta_w \log RV_t^{(w)} + \beta_m \log RV_t^{(m)} + u_{t+1}, \quad (7)$$

where $RV_t^{(d)}$, $RV_t^{(w)}$, and $RV_t^{(m)}$ denote daily, weekly, and monthly realized-volatility components. The HAR model proved remarkably successful empirically because it captures many observed features of long-memory volatility dynamics while remaining computationally simple and economically interpretable. The framework is closely connected to heterogeneous-agent theories in which market participants operate at different investment horizons (Müller et al., 1997). Daily components capture short-horizon trading activity, whereas weekly and monthly components reflect slower-moving institutional and macroeconomic adjustments.

The HAR framework also became a dominant benchmark in volatility forecasting because of its strong empirical performance relative to more complex nonlinear specifications. In the present paper, HAR models serve as the baseline forecasting architecture against which persistence-augmented models are evaluated.

Rough Volatility

The rough-volatility literature introduced a complementary perspective on volatility dynamics by emphasizing local path irregularity rather than long-horizon persistence. Gatheral et al. (2018) showed that volatility paths estimated from high-frequency financial data often display substantially rougher behavior than implied by standard Brownian-motion models. In particular, estimated Hurst exponents frequently lie well below the classical benchmark $H = \frac{1}{2}$.

A simplified rough-volatility representation may be written as

$$\log \sigma_t = \mu + \nu B_t^H, \quad (8)$$

where B_t^H denotes fractional Brownian motion with Hurst exponent $H < \frac{1}{2}$. From an econometric standpoint, roughness is commonly characterized through scaling relations involving moments of volatility increments,

$$E|\log \sigma_{t+\Delta}^2 - \log \sigma_t^2|^q \propto \Delta^{qH}, \quad (9)$$

where H governs the scaling behavior of local volatility fluctuations across short horizons. We estimate H_t by the slope of $\log m(q, \Delta)$ on $\log \Delta$ over a small grid of lags, divided by q .

The rough-volatility framework proved highly successful in explaining several empirical features of financial markets, including short-horizon volatility behavior and implied-volatility surface dynamics (Bennedsen et al., 2021). Rough models also provided improved consistency with observed option-market data and generated substantial interest in both academic and practitioner communities.

Importantly, roughness and long memory characterize different temporal properties of volatility. Roughness concerns highly local short-term irregularity, whereas long memory concerns persistence across broader temporal horizons. A volatility process may therefore be simultaneously rough and persistent. The empirical framework adopted in this paper explicitly incorporates both dimensions through rolling estimates of long-memory parameters and rolling Hurst exponents.

Semiparametric Estimation of Long Memory

Empirical estimation of long memory presents substantial econometric challenges because dependence structures must be distinguished from nonstationarity, regime shifts, and structural breaks. The literature therefore developed several semiparametric estimation procedures that focus specifically on low-frequency behavior.

One widely used approach is the Geweke–Porter–Hudak (GPH) log-periodogram estimator (Geweke and Porter-Hudak, 1983). The estimator exploits the low-frequency behavior of the spectral density by regressing the logarithm of the periodogram on transformed Fourier frequencies. Using the first m low-frequency ordinates, the estimator takes the form

$$\log I(\lambda_j) = \alpha - d \log[4 \sin^2(\lambda_j/2)] + u_j, \quad (10)$$

where $\hat{d} = -\hat{\beta}/2$ and bandwidth $m = \lfloor T^{0.65} \rfloor$.

Another important procedure is the local-Whittle estimator proposed by Robinson (1995). Rather than relying on regression, the local-Whittle approach minimizes an approximation to the Gaussian likelihood over the low-frequency band of the spectrum. The estimator generally exhibits lower asymptotic variance under broad conditions and has become one of the standard semiparametric procedures in long-memory econometrics. Both estimators focus specifically on low-frequency dependence without imposing complete parametric ARMA or GARCH structures, making them attractive for empirical applications in which the short-memory component may be misspecified or unstable.

In addition to semiparametric methods, the present paper also uses FIGARCH estimates on representative subsets of stocks as parametric cross-checks. Combining semiparametric and parametric approaches helps reduce dependence on any single modeling framework and improves robustness of the empirical conclusions.

Identification Challenges

Several identification problems complicate empirical interpretation of persistence estimates. The first concerns structural nonstationarity. Repeated regime shifts, evolving volatility states, and structural breaks can generate empirical behavior resembling fractional dependence even when the underlying process is not truly fractionally integrated (Mikosch and Stărică, 2004). As a result, estimated persistence parameters should not automatically be interpreted as evidence for stationary long-memory laws.

The second challenge concerns the distinction between highly persistent GARCH systems and genuine fractional integration. Near-integrated GARCH models may produce empirical dynamics that are difficult to distinguish from FIGARCH processes in finite samples, especially when persistence parameters approach unity. A third challenge concerns rough-volatility estimation. High-frequency financial data contain substantial market microstructure noise arising from bid-ask bounce, asynchronous trading, discreteness effects, and liquidity frictions. Unless realized-volatility measures are carefully constructed, estimated roughness parameters may partly reflect measurement distortions rather than genuine volatility dynamics.

The present paper addresses these issues through several robustness procedures. Persistence estimates are computed using multiple volatility proxies, including squared returns, Parkinson range-based measures, and realized variance where available. Both GPH and local-Whittle estimators are employed, together

with FIGARCH cross-checks and rolling estimation windows of varying lengths. The objective is not to establish exact structural identification of fractional stochastic laws, but rather to determine whether persistence-related measures behave coherently across market states and contain useful forecasting information.

These identification issues are central to the broader interpretation developed in the next section, where persistence measures are viewed as reduced-form indicators of uncertainty propagation rather than as mechanically interpreted fractional coefficients.

Structural Interpretation of Memory Parameters

The econometric literature traditionally introduces the memory parameter d as a statistical descriptor governing the decay rate of temporal dependence. Within purely probabilistic frameworks, d determines how strongly past shocks continue to influence present observations. Once one moves from abstract time-series theory to financial economics, however, this interpretation becomes incomplete. Financial markets are institutional systems in which information is generated, interpreted, delayed, transmitted, amplified, and sometimes distorted by heterogeneous participants. Persistence is not merely a property of a covariance function. It is an empirical manifestation of how long the consequences of information shocks remain economically active within the financial system.

This paper therefore interprets the rolling persistence estimate $\hat{d}_t$ as a reduced-form indicator of the expected duration of volatility shocks. More specifically, $\hat{d}_t$ summarizes the speed at which financial markets absorb informational disturbances and gradually return toward normal uncertainty conditions. Persistence measures should not be interpreted as identifying a single structural mechanism. Instead, they are treated as empirical state variables reflecting the combined influence of several interacting economic processes.

It is useful to distinguish carefully between statistical persistence and economic persistence. Statistical persistence refers to the slow decay of dependence in an observed series. Economic persistence refers to the prolonged influence of shocks on market uncertainty, liquidity conditions, risk perception, and portfolio adjustment. A macroeconomic announcement, credit event, or geopolitical disturbance may occur at a single point in time, yet its economic consequences may unfold gradually through changes in expectations, portfolio rebalancing, collateral conditions, funding constraints, and hedging demand. In this sense, volatility persistence reflects not only the arrival of information but also the prolonged process through which uncertainty is absorbed and resolved.

One important economic channel underlying persistence is information diffusion. Financial markets process information across multiple temporal horizons rather than through instantaneous adjustment. Even when prices respond rapidly to public news, the implications of uncertainty may continue to propagate through markets for extended periods. Market participants differ substantially in sophistication, leverage, liquidity access, and trading horizon. Consequently, adjustment to new information often occurs sequentially rather than simultaneously.

A second channel involves heterogeneous market participation. Financial systems contain agents operating at different frequencies and under different institutional constraints. High-frequency traders respond to order flow and liquidity conditions within milliseconds, whereas institutional investors, pension funds, and asset managers

may rebalance positions over substantially longer horizons. The aggregation of heterogeneous reactions across trading horizons can generate persistent volatility dynamics even when individual participants themselves do not follow fractional stochastic laws (Müller et al., 1997; Corsi, 2009). Under this interpretation, long memory represents the macroeconomic signature of micro-level heterogeneity.

Liquidity conditions provide a third channel through which persistence may arise. In highly liquid markets, shocks can often be absorbed relatively quickly because market depth and risk-sharing capacity remain strong. In stressed or illiquid markets, however, order imbalances may persist for extended periods, spreads may widen, and volatility may remain elevated long after the original informational disturbance occurs. Persistence measures may therefore capture not only informational dynamics but also the market's capacity to absorb risk.

Systemic stress and financial contagion constitute another important mechanism. Disturbances originating in one market segment may propagate gradually through collateral constraints, leverage adjustments, funding markets, and cross-asset portfolio exposures. Under such conditions, persistence reflects the duration of system-wide stress propagation rather than isolated asset-specific volatility dynamics. This interpretation is particularly relevant during major crisis periods such as the global financial crisis and the COVID episode, when uncertainty remained elevated across broad segments of the financial system for sustained periods.

Behavioral considerations also provide plausible explanations for persistent volatility dynamics. Financial-market participants may exhibit underreaction, delayed belief updating, ambiguity aversion, limited attention, or gradual expectation adjustment. Under such conditions, informational shocks are not fully incorporated into market behavior immediately. Instead, volatility may remain elevated while beliefs and positions adjust gradually over time.

These considerations motivate interpreting persistence measures as state-dependent indicators of the temporal organization of uncertainty in financial markets. A useful stylized representation may be written as

$$I_t = \sum_{k=0}^{\infty} w_k \varepsilon_{t-k}, \quad w_k \sim c k^{d-1}, \quad (11)$$

where ε_{t-k} denotes past informational shocks. The parameter d governs the speed at which the influence of past disturbances declines over time. Larger values of d correspond to slower decay and therefore longer persistence of uncertainty conditions.

Importantly, this interpretation remains intentionally reduced-form. The paper does not claim that persistence estimates uniquely identify a single economic mechanism. Persistence measures may simultaneously reflect heterogeneous trading horizons, delayed information diffusion, liquidity conditions, structural breaks, behavioral frictions, and systemic stress propagation. Their usefulness lies not in exact structural identification but in their ability to summarize economically meaningful temporal characteristics of financial uncertainty.

This reduced-form interpretation is conceptually similar to the interpretation of credit spreads in fixed-income markets. Credit spreads do not isolate a single risk component but instead summarize several overlapping influences, including default risk, liquidity premia, macroeconomic uncertainty, and risk aversion. Likewise,

persistence measures summarize several interacting features of the financial information environment rather than isolating one uniquely identifiable mechanism.

The empirical results developed later in the paper strongly support this interpretation. The cross-sectional mean of rolling persistence estimates rises sharply during major stress episodes, including the global financial crisis and COVID. Moreover, persistence measures display substantial positive association with the VIX and improve volatility forecasts particularly during high-stress regimes and at longer forecast horizons. These patterns are precisely what one would expect if persistence measures capture the expected duration of unresolved uncertainty.

The interpretation developed here also helps clarify the relationship between persistence and rough volatility. Roughness and long memory characterize different temporal properties of volatility dynamics rather than competing mechanisms. Roughness concerns local irregularity and short-horizon path behavior, whereas persistence concerns the prolonged influence of shocks across broader horizons. Financial markets may therefore exhibit sharp local volatility fluctuations immediately after news arrivals while simultaneously displaying prolonged uncertainty propagation over longer periods.

This distinction is especially important for volatility forecasting. Short-term volatility spikes and long-term persistence dynamics may contain different types of predictive information. Conventional volatility-level indicators such as the VIX primarily summarize the intensity of contemporaneous uncertainty. Persistence measures, by contrast, may summarize the expected duration of uncertainty conditions. Two periods with similar volatility levels may nevertheless differ substantially in how long elevated uncertainty is expected to persist.

From a forecasting perspective, persistence measures therefore provide information distinct from traditional volatility proxies. A high-persistence state implies not merely elevated volatility but also slow expected resolution of uncertainty. This interpretation motivates the forecasting framework developed in the next section, where persistence-related variables are incorporated directly into structured volatility forecasting systems.

Structured Forecasting with Persistence-Based Features

If persistence measures contain economically meaningful information, the next question concerns how such information should be incorporated into volatility forecasting systems. The approach adopted in this paper treats persistence measures not as isolated econometric coefficients but as dynamic state variables describing the temporal structure of financial uncertainty. Forecasting systems are therefore constructed around economically interpretable persistence-based features rather than around purely mechanical lag structures.

The framework developed here occupies an intermediate position between classical econometric modeling and purely data-driven machine learning. Traditional parametric forecasting systems provide interpretability and structural discipline but may fail to capture nonlinear interactions and regime dependence. Purely flexible machine-learning systems may capture complex patterns while sacrificing economic transparency and out-of-sample stability (Gu et al., 2020). The objective of the present framework is therefore not

to replace econometric structure with machine learning, but rather to combine economically interpretable feature construction with flexible forecasting architectures.

The central idea is that persistence contains information distinct from conventional volatility-level indicators. Traditional forecasting systems such as HAR models primarily capture the intensity and temporal aggregation of volatility. Market-state variables such as the VIX and MOVE indices summarize contemporaneous stress conditions. Persistence measures potentially contribute an additional dimension: the expected duration of uncertainty propagation.

This distinction is economically important. Two periods may exhibit similar contemporaneous volatility levels while differing substantially in how long elevated uncertainty is expected to persist. One period may involve a short-lived volatility spike associated with rapid uncertainty resolution, whereas another may involve prolonged stress propagation across sectors, funding markets, and institutional balance sheets. Persistence measures are designed to capture this second dimension.

The forecasting framework therefore decomposes volatility dynamics into three conceptually distinct channels. The first is the time-aggregation channel captured by HAR-type dynamics. The second is the contemporaneous stress-level channel captured by implied-volatility indicators such as VIX and MOVE. The third is the uncertainty-duration channel captured by persistence-related variables and persistence-by-stress interactions.

Formally, let $y_{t,h}$ denote the future volatility target at forecast horizon h , let $\mathbf{X}_t$ denote a vector of conventional predictors, and let $\mathbf{Z}_t$ denote a vector of persistence-based features. The forecasting problem may be written as

$$y_{t,h} = F(\mathbf{X}_t, \mathbf{Z}_t) + u_{t+h}, \quad (12)$$

where $F(\cdot)$ may represent either a linear or nonlinear forecasting system. The substantive contribution of the framework lies less in the specific estimator than in the construction of the persistence feature space. The feature vector includes several dimensions of persistence behavior:

$$\mathbf{Z}_t = \left(\hat{d}_t, \Delta \hat{d}_t, \text{Vol}(\hat{d})_t, \text{Trend}(\hat{d})_t, H_t, \Delta H_t, \bar{d}_t, \sigma_d^t, \bar{d}_{s(i),t}, \hat{d}_t \cdot \text{VIX}_t, \hat{d}_t \cdot \text{MOVE}_t \right). \quad (13)$$

These variables summarize persistence dynamics at several levels. The own-stock persistence estimate $\hat{d}_t$ captures local persistence behavior. Cross-sectional aggregates $\bar{d}_t$ and σ_d^t summarize system-wide persistence conditions, while sectoral averages $\bar{d}_{s(i),t}$ capture industry-specific uncertainty propagation. Interaction terms such as $\hat{d}_t \cdot \text{VIX}_t$ allow the impact of stress to depend on the persistence state of the market.

The forecasting architecture is organized as a layered ladder of models designed to isolate the marginal contribution of different persistence components. The baseline model is the standard HAR specification:

$$\textbf{Model A:} \quad \log \text{RV}_t^{(d)}, \log \text{RV}_t^{(w)}, \log \text{RV}_t^{(m)}, r_{t-1}, |r_{t-1}|.$$

The first augmentation introduces market-state variables:

$$\textbf{Model A}_1: \quad A + (\text{VIX}_t, \text{MOVE}_t).$$

This HAR-X structure measures the contribution of contemporaneous market stress independently of persistence effects.

The second augmentation introduces own-stock persistence and roughness features:

$$\textbf{Model A}_2: \quad A + (\hat{d}_t, \Delta \hat{d}_t, \text{Vol}(\hat{d})_t, \text{Trend}(\hat{d})_t, H_t, \Delta H_t).$$

The third and fourth augmentations isolate cross-sectional and sectoral persistence effects:

$$\textbf{Model A}_3: \quad A + (\bar{d}_t, \sigma_d^t),$$

$$\textbf{Model A}_4: \quad A + \bar{d}_{s(i),t}.$$

The fifth augmentation introduces persistence-by-stress interactions:

$$\textbf{Model A}_5: \quad A + (\hat{d}_t, \text{VIX}_t, \text{MOVE}_t, \hat{d}_t \cdot \text{VIX}_t, \hat{d}_t \cdot \text{MOVE}_t).$$

The complete structural forecasting system combines all persistence blocks simultaneously:

$$\textbf{Model C:} \quad A + A_2 + A_3 + A_4 + A_5.$$

Finally, the same feature space is estimated using several machine-learning procedures:

Model D: Lasso, Ridge, Elastic Net, Random Forest, Gradient Boosting.

This layered design is important because it isolates the marginal contribution of each persistence block relative to the HAR benchmark. The framework therefore allows one to determine whether predictive improvements arise primarily from market-state variables, own-stock persistence, cross-sectional persistence, sectoral aggregation, or persistence-by-stress interactions.

The architecture also allows evaluation of an important methodological question: whether forecasting gains arise primarily from economically structured feature design or from estimator flexibility itself. By comparing linear persistence-based models with flexible machine-learning estimators using the identical feature set, the framework isolates the incremental contribution of nonlinear estimation methods separately from the contribution of economically interpretable persistence features.

Several implementation safeguards are required because persistence-based variables are estimated quantities rather than directly observed variables. Rolling estimation windows are constructed to avoid look-ahead bias and to preserve the forecasting information set. Feature standardization is performed using training-sample information only. Hyperparameters are selected through time-series cross-validation rather than random cross-sectional shuffling. These precautions are particularly important because volatility forecasting systems are highly sensitive to temporal dependence structures.

Forecast evaluation also requires careful treatment because volatility proxies themselves contain measurement noise. In addition to mean squared error on the logarithmic realized-volatility scale, the framework therefore employs the QLIKE loss function proposed by Patton (2011), which is more robust when realized-volatility proxies contain observational error. For pairwise forecast comparisons, the paper uses the Diebold–Mariano statistic of Diebold and Mariano (1995) with the Harvey et al. (1997) finite-sample correction (HLN), computed on the cross-sectional mean loss differential per date

and using a Newey–West HAC variance with bandwidth $\lceil h/5 \rceil - 1$. Importantly, inference is conducted using panel-aware cross-sectional aggregation rather than naive pooled-cell independence assumptions. This distinction is critical because ignoring within-date cross-sectional dependence substantially inflates apparent statistical significance.

The forecasting structure developed here is therefore designed not merely to maximize predictive accuracy, but to investigate whether persistence measures behave like economically meaningful state variables. If persistence-based features systematically improve forecasts during prolonged stress episodes, particularly at longer horizons, then the interpretation of persistence as an indicator of uncertainty duration gains substantial empirical support.

As shown later in the empirical analysis, this is precisely what occurs. Persistence-related variables provide their largest incremental forecasting value during high-stress regimes and at longer forecast horizons, exactly where the duration of uncertainty becomes economically most important.

Data and Empirical Design

This section describes the data architecture, volatility construction procedures, rolling estimation framework, and forecasting protocol used in the empirical analysis. The central methodological principle is that persistence and roughness measures must be estimated in a manner consistent with the temporal structure of financial data while avoiding forward-looking contamination and preserving comparability across forecasting models.

The empirical framework is designed around three objectives. First, the data environment must be sufficiently long to capture multiple volatility regimes, including calm periods, financial crises, and post-crisis transitions. Second, the volatility proxies must be constructed carefully enough to distinguish genuine persistence from measurement noise. Third, the forecasting design must permit rigorous out-of-sample evaluation under realistic financial forecasting conditions.

Sample

The primary dataset consists of common equities drawn from the S&P 500 index with sufficiently long daily histories to support rolling estimation of persistence and roughness measures. Daily open, high, low, close, volume, and market-capitalization data are obtained from Bloomberg for an initial universe of 125 securities. After imposing a 70% data-coverage requirement over the sample period, the final balanced panel contains 115 stocks observed over 6,136 trading days from November 29, 2001 through April 21, 2026.¹

The sample spans several major financial regimes, including the post-2001 recovery period, the credit expansion preceding the global financial crisis, the global financial crisis itself, the European sovereign-debt episode, the long low-volatility post-crisis expansion, the COVID shock, and the post-COVID inflationary environment. The presence of these distinct regimes is essential because the paper interprets persistence measures as state-dependent indicators of uncertainty propagation.

In addition to stock-level data, the analysis incorporates several market-wide variables intended to capture broad financial conditions.

These include the S&P 500 index, Dow Jones Industrial Average, Nasdaq 100, Russell 2000, the VIX implied-volatility index, the MOVE Treasury-volatility index, Treasury-yield series, the 2s/10s yield-curve slope, and investment-grade and high-yield CDX credit indices. Sector-level aggregation is implemented using GICS Level-1 classifications together with SPDR sector ETFs.

To supplement the daily data, the study also incorporates a smaller high-frequency subsample consisting of five-minute observations for the S&P 500 and twenty liquid equities over the period October 2025 through April 2026. These data are used primarily for robustness analysis involving realized-volatility estimation and rough-volatility diagnostics rather than as the primary forecasting sample.

The resulting dataset provides several important advantages. The long sample horizon allows persistence measures to be evaluated across multiple crisis periods and volatility states. The cross-sectional structure permits construction of aggregate persistence measures and sectoral persistence indices. The high-frequency subsample allows comparison between daily volatility proxies and realized-volatility estimators.

Volatility Proxies

Daily logarithmic returns are computed from adjusted closing prices and winsorized at the 0.1% and 99.9% percentiles within each stock in order to reduce the influence of recording errors, corporate actions, and isolated outlier observations. Because volatility measurement plays a central role in persistence estimation, the choice of volatility proxy is particularly important.

The primary volatility measure used throughout the paper is the Parkinson range-based variance estimator,

$$\text{RVPK}_t = \frac{(\ln H_t - \ln L_t)^2}{4 \ln 2}, \quad (14)$$

where H_t and L_t denote the daily high and low prices. Unlike squared close-to-close returns, the Parkinson estimator exploits intraday price range information and therefore provides a more efficient volatility proxy under continuous-diffusion assumptions. The Parkinson measure offers several advantages for the present application. First, it is available consistently over the entire 25-year sample period for all stocks in the panel. Second, it substantially reduces measurement noise relative to squared returns. Third, the estimator captures volatility clustering particularly clearly, which is important for persistence estimation.

The empirical analysis also employs several alternative volatility proxies for robustness purposes, including squared returns, absolute returns, and realized variance constructed from high-frequency data where available. These alternative measures help determine whether the persistence results depend critically on the choice of volatility proxy.

For forecasting purposes, the volatility target for stock i at horizon h is defined as the logarithm of future average realized variance,

$$y_{i,t,h} = \log\left(\frac{1}{2} h \sum_{j=1}^h \text{RVPK}_{i,t+j}\right). \quad (15)$$

Three forecast horizons are considered, $h \in \{1, 5, 22\}$, corresponding approximately to daily, weekly, and monthly horizons. This multi-horizon structure is important because the economic interpretation of persistence predicts that persistence-related features should

¹ Accessed: 21 April 2026 , Bloomberg Terminal.

contribute most strongly at longer horizons where uncertainty-duration effects become more relevant.

Descriptive Statistics

Table 2 reports summary statistics for daily log returns pooled across the 115-stock panel and broken down by GICS sector. The pooled annualized mean return equals approximately 9.4%, while the pooled annualized standard deviation equals approximately 31.5%.

More importantly, the distribution of returns displays substantial non-normality. The pooled excess kurtosis exceeds 11, and the Jarque–Bera statistic overwhelmingly rejects Gaussianity. A fitted Student-*t* distribution implies degrees of freedom near 2.7, indicating extremely heavy tails. Substantial cross-sectional heterogeneity is also present across sectors. Information Technology and Financials display the largest annualized volatility levels, whereas Utilities and Consumer Staples display substantially lower volatility. Real Estate and Financials exhibit especially heavy-tailed behavior, consistent with their sensitivity to systemic financial conditions and leverage dynamics.

Table 1 Data Description

Panel A: Sample Composition by GICS Sector		
Sector	N Stocks	% of Sample
Financials	19	16.5%
Information Technology	18	15.7%
Industrials	17	14.8%
Health Care	15	13.0%
Consumer Staples	11	9.6%
Consumer Discretionary	9	7.8%
Real Estate	7	6.1%
Utilities	6	5.2%
Communication Services	5	4.3%
Energy	5	4.3%
Materials	3	2.6%
Total	115	100.0%

Panel B: Market-Level Variables					
Variable	Mean	Std	Min	Max	N
SPX	2398.64	1555.37	676.53	7126.06	6136
INDU	19990.01	11002.39	6547.05	50188.14	6136
NDX	6405.01	6378.05	804.64	26672.43	6136
RTY	1183.03	601.33	327.09	2792.96	6136
VIX	19.48	8.45	9.14	82.69	6136
MOVE	88.44	31.15	36.62	264.60	6136
USGG3M	1.71	1.83	-0.14	5.51	6136
USGG10YR	3.15	1.16	0.51	5.43	6136
USYC2Y10	110.42	96.62	-108.71	291.03	6136
CDX IG 5Y	69.62	19.05	43.75	151.80	3674
CDX HY 5Y	104.59	4.19	86.25	110.41	3669

Notes: Sample period Nov 2001 – Apr 2026. Market-level variables sourced from Bloomberg.

Table 1 summarizes the composition of the panel together with the distribution of market-wide state variables. The VIX ranges from approximately 9 to above 82, while the MOVE index ranges from approximately 37 to above 264. These substantial variations in market stress are particularly important because the structural

Table 2 Summary Statistics for Daily Stock Returns

	Mean (% ann.)	Std (% ann.)	Skewness	Kurtosis	Min (%)	Max (%)
Panel A: Pooled Sample All Stocks (N=115)	9.38	31.48	-0.13	11.64	-25.60	26.73
Panel B: By GICS Sector						
Communication Services (n=5)	12.45	32.90	-0.07	6.40	-12.67	11.90
Consumer Discretionary (n=9)	12.53	30.26	-0.05	5.91	-11.45	10.87
Consumer Staples (n=11)	8.23	22.78	-0.15	6.33	-8.95	8.48
Energy (n=5)	8.59	30.83	-0.24	5.82	-11.72	11.25
Financials (n=19)	6.74	34.38	-0.14	11.14	-15.00	14.85
Health Care (n=15)	9.09	27.95	-0.17	6.27	-10.79	10.12
Industrials (n=17)	9.37	27.28	-0.19	5.17	-9.96	9.28
Information Technology (n=18)	11.41	36.16	-0.13	5.22	-13.04	12.17
Materials (n=3)	9.29	33.39	-0.34	5.32	-12.67	10.66
Real Estate (n=7)	11.80	34.15	0.11	11.17	-13.87	15.37
Utilities (n=6)	6.35	22.39	-0.12	7.65	-8.89	9.57

Notes: Sample period Nov 2001 – Apr 2026, 115 S&P 500 constituents that pass a 70% coverage filter. Mean and Std are annualized; Skewness and Kurtosis are excess values. Returns are log returns winsorized at the 0.1% and 99.9% percentiles.

interpretation of persistence predicts that persistence measures should rise systematically during periods of elevated uncertainty.

The descriptive evidence therefore already suggests several key themes that are explored later in the forecasting analysis. Financial returns are highly non-Gaussian, volatility dynamics exhibit substantial persistence, and market-wide stress conditions vary dramatically across the sample period.

Preliminary

Before proceeding to rolling persistence estimation and forecasting analysis, it is important to examine several basic empirical properties of the data. The preliminary diagnostics confirm the presence of the central stylized facts motivating the paper: volatility clustering, strong nonlinear dependence in volatility measures, substantial heavy-tailed behavior, cross-sectional comovement, and persistent conditional heteroskedasticity.

Figure 1 presents four preliminary diagnostics for a representative stock, together with the VIX index. Panel (a) displays daily logarithmic returns for AAPL, while panel (b) reports the corresponding Parkinson range-based variance proxy. The contrast between the two series immediately reveals one of the classical stylized facts of financial econometrics. Returns themselves fluctuate around zero with little visually detectable serial dependence, whereas volatility measures display pronounced clustering. Large volatility episodes tend to be followed by additional large volatility episodes, while calm periods tend to persist as well.

The recession-shaded regions corresponding to the global financial crisis and COVID are especially informative. During these periods, the Parkinson variance proxy rises sharply and remains elevated for extended intervals rather than reverting immediately to calm-state levels. This behavior is precisely the type of prolonged uncertainty propagation that motivates the interpretation of persistence measures as indicators of stress duration rather than merely volatility magnitude.

Panel (c) of Figure 1 compares autocorrelation functions for raw returns, squared returns, and Parkinson realized variance. The results are striking. Raw returns exhibit essentially no significant serial dependence beyond the first few lags, consistent with the approximate unpredictability of returns documented throughout the financial-econometrics literature. By contrast, both squared returns and the Parkinson variance proxy display autocorrelations that decay slowly and remain significantly positive far beyond short horizons. The Parkinson autocorrelations lie consistently above those for squared returns, suggesting that the range-based estimator provides a less noisy volatility proxy.

This distinction is important because persistence estimation is highly sensitive to measurement error. Squared returns contain substantial noise arising from isolated jumps and market microstructure effects. By exploiting intraday high–low information, the Parkinson estimator substantially improves the signal-to-noise ratio of volatility measurement.

Panel (d) presents the VIX time series over the full sample period. The VIX displays several dramatic spikes corresponding to major stress episodes, particularly the global financial crisis and COVID. These broad market stress indicators later play a central role in the persistence-by-stress interaction framework developed in Section 6.

Figure 2 extends the diagnostic analysis in several directions. Panel (a) compares the empirical pooled distribution of daily returns

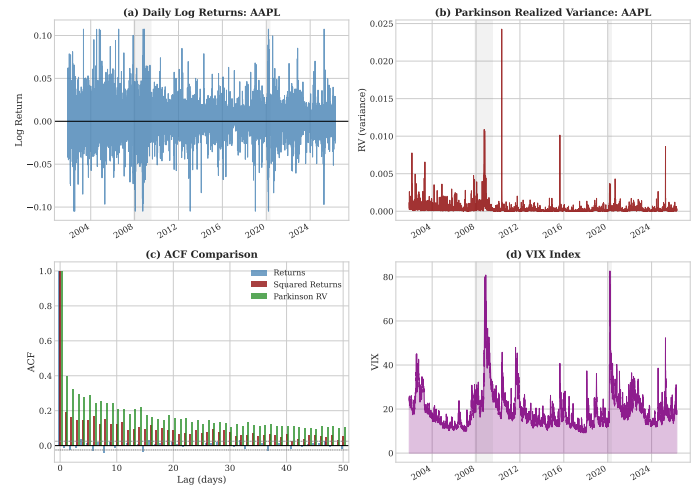


Figure 1 Daily log returns and Parkinson range-based variance proxy for a representative stock; ACF comparison; and the VIX time series. Recession-shaded windows mark the global financial crisis and COVID. *Takeaway: volatility clusters visibly; squared returns and RVPK exhibit slow autocorrelation decay out to 50 lags while raw returns do not.*

with fitted Gaussian and Student- t distributions. The empirical distribution displays substantially heavier tails than the Gaussian benchmark, particularly in the extremes. The associated Jarque–Bera statistic overwhelmingly rejects normality, while the QQ-plot in panel (d) displays the classical S-shaped departure from Gaussian behavior characteristic of financial return distributions. Extreme observations occur far more frequently than predicted by normal-diffusion models. This heavy-tailed structure has important implications for volatility modeling because persistence estimation can be distorted when tail behavior is ignored.

Panel (b) of Figure 2 reports cross-stock Parkinson variance correlations for ten representative equities, ranging from 0.16 (KO–AAPL) to 0.85 (PG–CL). The heatmap reveals substantial cross-sectional heterogeneity in volatility comovement, with particularly large correlations among firms exposed to similar macroeconomic or sectoral conditions. These cross-sectional relationships motivate the construction of aggregate persistence variables later in the forecasting framework. If uncertainty propagates systemically across sectors and balance sheets, then cross-sectional persistence aggregates should contain information beyond individual-stock volatility dynamics.

Panel (c) compares FIGARCH conditional volatility estimates with rolling Parkinson realized volatility for AAPL. The two series track each other closely throughout the sample, including major stress periods. This result provides an important preliminary validation of the fractional-volatility framework. Table 3 gives FIGARCH point estimates for ten stocks with normal innovations: the mean estimate is $\hat{d} = 0.329$ with values in $[0.237, 0.461]$, all within the stationary long-memory range $(0, \frac{1}{2})$ and consistent with the semiparametric estimates we report below.

Taken together, the preliminary diagnostics establish several important empirical conclusions. First, volatility measures exhibit strong persistence while returns themselves remain approximately serially uncorrelated. Second, financial returns are substantially heavy-tailed and strongly non-Gaussian. Third, volatility dynamics

display substantial cross-sectional comovement. Fourth, long-memory volatility models provide empirically plausible descriptions of observed volatility behavior. Most importantly, the diagnostics strongly support the interpretation that volatility persistence is not an isolated feature of a few securities or crisis episodes. Rather, persistent volatility dynamics appear broadly throughout the cross-section and evolve systematically across market states.

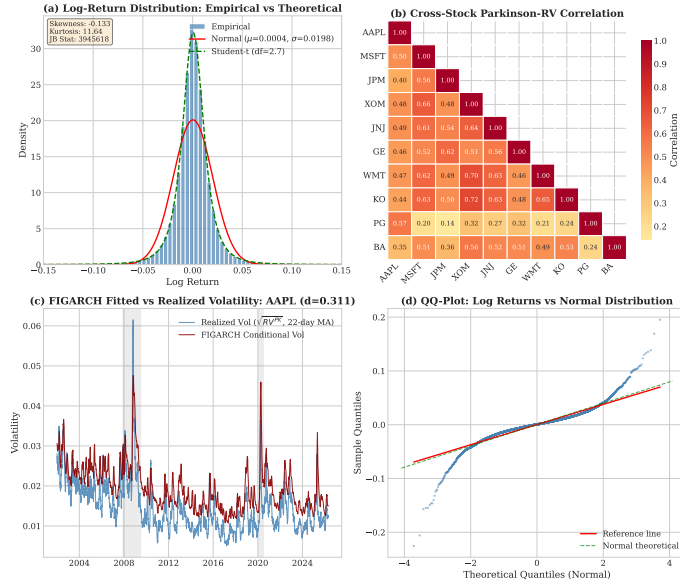


Figure 2 Preliminary diagnostics. (a) Pooled log-return distribution with Normal and Student- t fits. (b) Cross-stock Parkinson-RV correlation heatmap. (c) FIGARCH(1, d ,1) conditional volatility versus realised volatility for AAPL. (d) QQ-plot against a normal reference. *Takeaway: returns are heavily fat-tailed; FIGARCH reproduces RVPK-scale volatility well across the full sample.*

Table 3 FIGARCH(1, d ,1) Model Estimates

Stock	ω	d	ϕ	β	AIC	BIC
AAPL	0.2153	0.311	0.313	0.524	25240.0	25280.3
MSFT	0.2259	0.309	0.218	0.426	22734.1	22774.4
JPM	0.1213	0.461	0.182	0.510	23429.5	23469.8
XOM	0.0671	0.421	0.231	0.557	21309.8	21350.2
JNJ	0.0725	0.308	0.346	0.498	17233.5	17273.8
GE	0.0904	0.378	0.311	0.556	23508.0	23548.3
WMT	0.1606	0.244	0.378	0.483	19624.8	19665.1
KO	0.1433	0.273	0.000	0.175	17667.8	17708.1
PG	0.1897	0.237	0.000	0.116	17635.3	17675.6
BA	0.1723	0.346	0.322	0.568	24599.0	24639.4
Mean	—	0.329	—	—	—	—

Notes: FIGARCH(1, d ,1) with AR(1) mean and normal innovations. $d \in (0, 0.5)$ indicates stationary long memory in conditional variance.

Rolling Estimation Design

Persistence and roughness measures are estimated within rolling windows so that they behave as dynamic state variables rather than fixed full-sample constants. This distinction is essential for the

interpretation developed throughout the paper. If persistence reflects evolving uncertainty conditions, information diffusion, and regime-dependent stress propagation, then persistence measures themselves must be allowed to vary over time.

For each stock i and each end date t , the fractional integration parameter $\hat{d}_{i,t}$ is estimated using a rolling 750-trading-day window. Both the Geweke–Porter–Hudak log-periodogram estimator and the local-Whittle semiparametric estimator are employed using bandwidth $m = \lfloor T^{0.65} \rfloor$. The rolling estimates are updated every five trading days, producing weekly time series of $\hat{d}_{i,t}^{GPH}$, $\hat{d}_{i,t}^{LW}$, and $H_{i,t}$, where $H_{i,t}$ denotes the rolling Hurst exponent estimated from logarithmic Parkinson realized variance.

The choice of weekly updating frequency is intentional. Daily updating would produce extremely noisy persistence estimates dominated by local estimation variation rather than economically meaningful state changes. Monthly updating, on the other hand, would smooth away important regime transitions. Weekly updating provides an intermediate compromise that remains responsive to changing market conditions while preserving reasonable statistical stability.

The 750-day estimation window is similarly motivated by the tradeoff between statistical precision and state sensitivity. Persistence estimation requires relatively long windows because long-memory parameters are fundamentally low-frequency objects. Very short windows generate unstable estimates with large sampling variance. Extremely long windows, however, blur structural regime changes and reduce responsiveness to evolving market conditions. The selected window length therefore balances these competing considerations.

The rolling framework transforms persistence from a static full-sample coefficient into a genuinely time-varying market-state variable. This transformation is central to the economic interpretation of the paper. Under the rolling design, persistence measures may rise during prolonged stress episodes and decline during calm periods, allowing direct analysis of uncertainty-duration dynamics.

From these rolling estimates, the paper constructs the persistence feature vector introduced in Section 6. The feature set includes several categories of variables. The first category consists of own-stock persistence dynamics,

$$\hat{d}_{i,t}, \Delta \hat{d}_{i,t}, \text{Vol}(\hat{d}_{i,t}), \text{Trend}(\hat{d}_{i,t}), H_{i,t}, \Delta H_{i,t}, \quad (16)$$

summarizing the level, evolution, variability, and trend structure of persistence and roughness at the individual-stock level. The second category consists of cross-sectional persistence aggregates,

$$\bar{d}_t, \sigma_d^t, \text{Skew}_t(d), \text{Kurt}_t(d), \quad (17)$$

which summarize system-wide persistence conditions. The cross-sectional mean $\bar{d}_t$ is especially important because the paper interprets it as a broad indicator of the duration of unresolved uncertainty in financial markets. The third category consists of sectoral persistence measures, $\bar{d}_{s(i),t}$, where the average is computed across stocks belonging to the same GICS sector as stock i , intended to capture industry-specific stress propagation and heterogeneous information environments.

Additional features include threshold indicators $\mathbf{1}_{\{\hat{d}_{i,t} > \tau\}}$ together with interaction terms involving VIX, MOVE, and liquidity

conditions. These interaction variables allow the forecasting impact of persistence to depend explicitly on market stress states.

Table 4 reports pooled descriptive statistics for the resulting feature set. Several patterns are immediately noteworthy. First, the local-Whittle estimates are systematically larger than the GPH estimates, consistent with earlier long-memory studies. Second, the Hurst estimates remain uniformly well below $\frac{1}{2}$, confirming rough-volatility behavior. Third, the cross-sectional mean persistence measure varies substantially over time, supporting its interpretation as a dynamic state variable rather than a stable structural constant.

Table 4 Persistence Feature Vector: Definitions and Pooled Statistics

Category	Feature	Mean	Std	1%	99%
LRD	$\hat{d}_{GPH,t}$ (rolling)	0.1731	0.0961	0.0008	0.4154
	$\hat{d}_{LW,t}$ (rolling)	0.3358	0.1738	0.0265	0.8278
Roughness	H_t (rolling)	0.0620	0.0275	0.0196	0.1303
Memory dyn.	$\Delta \hat{d}_t$	-0.0001	0.0150	-0.0478	0.0478
	$\text{Vol}(\hat{d})_t$	0.0067	0.0112	0.0000	0.0587
	$\text{Trend}(\hat{d})_t$	-0.0001	0.0080	-0.0266	0.0252
HAR	RV^d	0.0003	0.0012	0.0000	0.0027
	RV^w	0.0003	0.0010	0.0000	0.0027
	RV^m	0.0003	0.0008	0.0000	0.0025
Sector aggregate	$\bar{d}_{s(i),t}$	0.1730	0.0803	0.0446	0.3362
Cross-sectional	$\bar{d}_t$ (cross-sectional mean)	0.1731	0.0746	0.0700	0.3029
	σ_d^t (cross-sectional std)	0.0596	0.0118	0.0335	0.0838
	Skew of d_t across stocks	0.0165	0.4559	-1.1219	0.8546
	Kurt of d_t across stocks	0.3385	0.8753	-0.7179	2.8986
	$P(\bar{d} > 0.30)$	0.1169	0.1747	0.0000	0.5217
	Range of $\bar{d}$ across stocks	0.3077	0.0695	0.1905	0.4992
Market	VIX	19.0494	8.3693	9.8977	53.7306
	MOVE	84.6779	30.7937	44.3771	188.2890
	USYC2Y10	96.3550	94.2156	-82.3963	280.2324
Interaction	$\hat{d}_t \cdot \text{VIX}_t$	3.6108	3.1447	0.0125	14.7306
	$\hat{d}_t \cdot \text{MOVE}_t$	15.3338	11.6301	0.0536	54.1197
	$\hat{d}_t / \text{Liq}_t$	0.0000	0.0000	0.0000	0.0000

Notes: Statistics are pooled across all stocks and dates in the rolling-estimation panel. $\hat{d}_t$ values are taken on a weekly stride; HAR components are constructed from Parkinson realised variance. Sector aggregates use GICS level-1; threshold indicators and interaction terms (with VIX, MOVE, and inverse dollar-volume liquidity) use the GPH estimate of $\hat{d}_t$.

The persistence-by-VIX interaction variable becomes very large during crisis periods, suggesting that persistence and market stress reinforce one another during prolonged uncertainty episodes.

An important feature of the rolling design is that all variables are constructed strictly using information available at time t . This prevents forward-looking contamination and ensures that the forecasting exercise represents a genuine out-of-sample evaluation

rather than a pseudo-exercise using future information implicitly embedded in the feature set.

The rolling framework also provides a natural bridge between econometric estimation and economic interpretation. Under the structural interpretation developed earlier, persistence should rise during periods in which uncertainty dissipates slowly and should decline during periods in which shocks are absorbed rapidly. The rolling estimates allow this prediction to be tested directly in the empirical analysis.

Moreover, because the rolling framework generates persistence measures for each stock, sector, and date simultaneously, it becomes possible to study not only average persistence but also the cross-sectional dispersion and heterogeneity of persistence conditions throughout the financial system. The resulting persistence-state panel therefore forms the core empirical object underlying the forecasting analysis developed in the next section.

Out-of-Sample Forecasting Protocol

The forecast evaluation is strictly walk-forward. We use the first 40% of the sample dates (sample dates 1 through 431, ending mid-2013) as the initial training window; thereafter we re-fit each model and produce a forecast at every subsequent sample date. For the linear ladder (Models $A, A_1, A_2, A_3, A_4, A_5, C$) we re-fit by ordinary least squares at every step (expanding window). For Model D we re-fit every twenty sample steps to keep the computational burden tractable for the tree-based estimators; results are not sensitive to this stride. All feature standardization, hyperparameter selection, and model selection are performed on training-window information only. Both the realized target and the forecast are recorded as $(T_{\text{eval}} \times N)$ panels, which then feed the loss functions, the Diebold–Mariano test, and the regime/sector splits in Section 8.

Numerical Analysis and Forecasting Results

This section presents the empirical results of the paper. The analysis proceeds in several stages. First, we examine full-sample estimates of long-memory and roughness parameters across the cross-section of equities. Second, we study the dynamics of rolling persistence measures across market regimes. Third, we evaluate the forecasting performance of the layered forecasting framework developed in Section 6. Finally, we investigate robustness across sectors, volatility states, liquidity conditions, and alternative estimation procedures.

The central empirical question is whether persistence measures behave like economically meaningful state variables rather than purely statistical nuisance parameters. Under the interpretation developed earlier, persistence measures should rise during sustained stress episodes, co-move with broad uncertainty indicators, and contribute forecasting information particularly during periods in which uncertainty resolves slowly.

Long-Memory and Roughness Estimates

We begin with full-sample cross-sectional estimates of persistence and roughness. The results strongly confirm the simultaneous presence of long-memory behavior and rough volatility across the equity panel.

The first important finding is that raw returns themselves display essentially no evidence of long memory. The cross-sectional mean GPH estimate for returns is approximately $\hat{d}_{\text{returns}}^{GPH} = -0.011$, with almost no stocks exhibiting statistical significance at conventional

Table 5 Long-Range Dependence and Roughness Estimates by Sector

Sector	Returns		Parkinson RV			Roughness	
	$\hat{d}_{GPH}$	$\hat{d}_{LW}$	$\hat{d}_{GPH}$	$\hat{d}_{LW}$	% Sig	$\bar{H}$	N
Communication Services	0.009	0.002	0.239	0.477	100%	0.076	5
Consumer Discretionary	-0.009	-0.028	0.227	0.444	100%	0.056	9
Consumer Staples	-0.018	-0.041	0.192	0.381	91%	0.059	11
Energy	-0.004	-0.030	0.208	0.439	100%	0.067	5
Financials	-0.013	-0.033	0.245	0.461	100%	0.071	19
Health Care	-0.017	-0.030	0.215	0.419	100%	0.058	15
Industrials	-0.005	-0.015	0.222	0.437	94%	0.060	17
Information Technology	-0.007	-0.017	0.234	0.455	100%	0.061	18
Materials	-0.010	-0.017	0.232	0.448	100%	0.061	3
Real Estate	-0.005	-0.037	0.255	0.486	100%	0.069	7
Utilities	-0.032	-0.066	0.200	0.404	100%	0.062	6
Overall	-0.011	-0.028	0.226	0.440	98%	0.063	115

Notes: d is the fractional differencing parameter. GPH is the Geweke–Porter–Hudak log-periodogram estimator with bandwidth $m = T^{0.65}$; LW is the local-Whittle semiparametric estimator. Returns are daily log returns; Parkinson RV is the range-based realized variance from daily high–low prices. H is the Hurst exponent of $\log RV^{PK}$ estimated from the scaling of the $q = 2$ moment of increments over lags $\{1, 2, 3, 5, 8, 13, 21\}$. $H < 0.5$ indicates rough behaviour. % Sig reports the share of stocks whose volatility $\hat{d}_{GPH}$ is significant at 5%.

levels. This result is consistent with the approximate unpredictability of financial returns documented throughout the empirical asset-pricing literature.

The situation changes dramatically once one examines volatility measures. For the Parkinson range-based variance proxy, the cross-sectional mean GPH estimate equals approximately $\hat{d}_{RV}^{GPH} = 0.226$, while the corresponding local-Whittle estimate equals approximately $\hat{d}_{RV}^{LW} = 0.440$. Moreover, between 98% and 100% of stocks exhibit statistically significant persistence estimates depending on the estimator employed. These results provide strong evidence that long-memory behavior is a broad characteristic of volatility dynamics rather than an isolated feature of a few securities.

The rough-volatility results are equally striking. The cross-sectional mean Hurst exponent estimated from logarithmic Parkinson realized variance equals approximately $H = 0.063$, with every stock in the panel satisfying $H < \frac{1}{2}$. This result implies that volatility is universally rough at short horizons throughout the equity panel. The cross-sectional dispersion of Hurst estimates is also remarkably small, suggesting that rough-volatility behavior is highly stable across sectors and firms.

Table 5 reports these estimates separately by GICS sector. Financials and Real Estate exhibit among the largest persistence estimates, consistent with their exposure to systemic stress propagation, leverage conditions, and liquidity cycles. Utilities and Consumer Staples display somewhat smaller persistence estimates, although long-memory behavior remains substantial throughout all sectors. An important feature of the results is the remarkable stability of persistence estimates across the cross-section: the standard deviation of persistence estimates across stocks remains relatively modest, indicating that long-memory volatility behavior is fundamentally a panel-wide phenomenon rather than the consequence of a few extreme outliers.

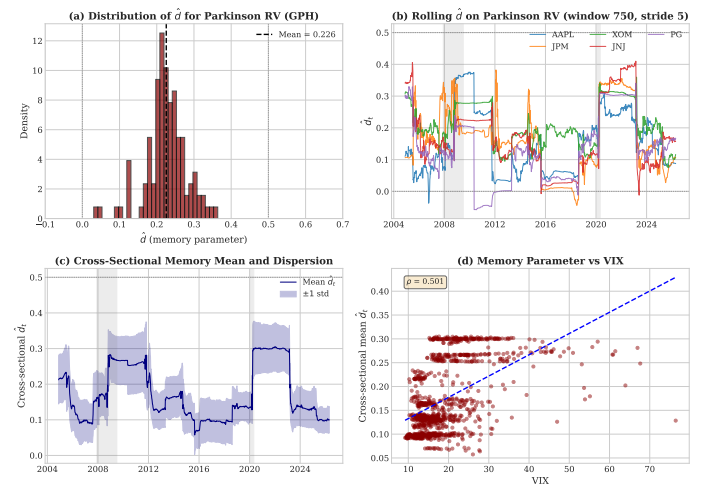


Figure 3 Rolling persistence estimates. (a) Cross-sectional distribution of full-sample GPH $\hat{d}$ on Parkinson range-based variance proxy. (b) Rolling $\hat{d}^{GPH}$ for AAPL, JPM, XOM, JNJ, PG. (c) Cross-sectional mean $\bar{d}_t$ with ± 1 standard deviation band. (d) $\bar{d}_t$ versus VIX, with linear fit and correlation. *Takeaway:* the cross-sectional mean memory parameter rises sharply in the GFC and COVID and tracks the VIX with correlation $\rho = +0.50$, consistent with its interpretation as a duration-of-stress indicator.

Figure 3 reports four time-series and cross-sectional views of the rolling estimation. Panel (a) reports the distribution of full-sample GPH estimates across all 115 stocks. The distribution is tightly concentrated around the interval $0.15 \leq \hat{d} \leq 0.30$, with relatively few extreme values.

Panel (b) reports rolling $\hat{d}^{GPH}$ for five representative stocks (AAPL, JPM, XOM, JNJ, PG). The most striking feature is the broad upward shift in persistence during major crisis episodes, particularly during the global financial crisis and COVID.

Panel (c) presents perhaps the most important empirical figure in the paper: the cross-sectional mean persistence state $\bar{d}_t = N^{-1} \sum_i \hat{d}_{i,t}$, together with a one-standard-deviation band. The dynamics of this aggregate persistence state are highly informative. During the calm 2013–2014 period, the cross-sectional mean

persistence measure remains near approximately $\bar{d}_t \approx 0.154$. During the global financial crisis, the cross-sectional mean rises to approximately $\bar{d}_t \approx 0.259$, representing an increase of approximately 68%. During COVID, the persistence state rises even further to approximately $\bar{d}_t \approx 0.287$, an increase of approximately 86% relative to the calm-state baseline.

These increases are economically large and strongly consistent with the interpretation of persistence as an indicator of uncertainty duration. Financial crises are not merely periods of elevated volatility. They are periods in which uncertainty remains unresolved for extended intervals. The persistence state appears to measure precisely this dimension.

Panel (d) of Figure 3 examines the relationship between the cross-sectional persistence state and the VIX. The estimated contemporaneous correlation equals approximately $\rho(\bar{d}_t, \text{VIX}_t) = 0.50$. This result is especially important because it demonstrates that persistence and contemporaneous market stress are closely related but not identical. The VIX primarily measures the intensity of current uncertainty, whereas persistence appears to measure the expected duration of uncertainty propagation. The distinction becomes clear during prolonged crisis periods. Elevated volatility may persist even after the initial shock has occurred because uncertainty continues to propagate gradually through balance sheets, funding conditions, leverage adjustments, and institutional portfolios.

Taken together, these findings provide direct empirical support for the reduced-form interpretation of Section 5. The rolling persistence state behaves like a dynamic indicator of how slowly the financial system is absorbing uncertainty. It rises during broad-based stress episodes and declines gradually as uncertainty dissipates. The next subsection investigates whether these persistence states also contain economically meaningful forecasting information.

Out-of-Sample Forecast Comparison

We now evaluate the forecasting performance of the layered forecasting framework introduced in Section 6. The analysis focuses on whether persistence-related features improve volatility forecasts beyond standard HAR and HAR-X benchmarks and, more importantly, whether such improvements occur precisely in the market states predicted by the interpretation developed earlier.

Table 6 reports the main out-of-sample forecasting results across the full forecasting ladder $A \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow A_4 \rightarrow A_5 \rightarrow C \rightarrow D$, evaluated at the daily, weekly, and monthly forecast horizons $h \in \{1, 5, 22\}$ and Figure 4 summarizes the layered ablation visually. The figure makes clear that most of the pooled forecasting improvement comes from market-state variables, while the additional persistence-related gains become most visible at the monthly horizon. Forecast performance is measured primarily through mean squared error on the logarithmic Parkinson realized-volatility scale together with the QLIKE loss function. Statistical significance is evaluated using the Harvey–Leybourne–Newbold finite-sample-corrected Diebold–Mariano statistic computed under panel-aware cross-sectional aggregation.

The HAR baseline (Model A) produces mean squared errors approximately equal to 0.669, 0.363, 0.269 for the daily, weekly, and monthly horizons respectively. We note in passing that the iid pooled-cell DM statistic ignores within-date cross-sectional dependence and

is inflated by roughly a factor of six relative to the panel-aware HLN-corrected version (e.g., +24.0 versus +3.87 for Model C at $h = 5$); we report the HLN values throughout.

Table 6 Out-of-Sample Forecast Comparison: Layered Ladder $A \rightarrow A_1 \dots A_5 \rightarrow C \rightarrow D$, Horizons $h \in \{1, 5, 22\}$

Model	h	MSE(log RV^{PK})	QLIKE	% Δ vs A	HLN DM- t vs A
A	1	0.6686	-7.5872	+0.00%	–
	5	0.3634	-7.6731	+0.00%	–
	22	0.2689	-7.6121	+0.00%	–
A1	1	0.6345	-7.6340	+5.10%	+5.34***
	5	0.3353	-7.6949	+7.72%	+5.31***
	22	0.2589	-7.6156	+3.72%	+1.98**
A2	1	0.6722	-7.5780	-0.54%	-3.06***
	5	0.3635	-7.6712	-0.03%	-0.07
	22	0.2697	-7.6083	-0.30%	-0.41
A3	1	0.6692	-7.5918	-0.10%	-0.58
	5	0.3621	-7.6755	+0.37%	+1.05
	22	0.2655	-7.6179	+1.26%	+1.41
A4	1	0.6689	-7.5843	-0.05%	-0.45
	5	0.3627	-7.6717	+0.21%	+0.80
	22	0.2681	-7.6108	+0.29%	+0.47
A5	1	0.6358	-7.6442	+4.91%	+4.02***
	5	0.3357	-7.6970	+7.63%	+4.18***
	22	0.2612	-7.6170	+2.87%	+1.38
C	1	0.6382	-7.6422	+4.55%	+3.15***
	5	0.3334	-7.6992	+8.24%	+3.87***
	22	0.2540	-7.6258	+5.54%	+2.33**
D.lasso	1	0.6403	-7.6316	+4.23%	+3.92***
	5	0.3381	-7.6940	+6.97%	+4.16***
	22	0.2680	-7.6113	+0.33%	+0.20
D.ridge	1	0.6452	-7.6301	+3.50%	+3.28***
	5	0.3418	-7.6931	+5.95%	+3.60***
	22	0.2698	-7.6121	-0.32%	-0.08
D.en	1	0.6405	-7.6305	+4.20%	+3.97***
	5	0.3382	-7.6939	+6.93%	+4.23***
	22	0.2677	-7.6117	+0.44%	+0.25
D.rf	1	0.6658	-7.6177	+0.42%	+0.41
	5	0.3553	-7.6824	+2.22%	+1.62
	22	0.2831	-7.6002	-5.29%	-2.24**
D.gbm	1	0.7584	-7.5710	-13.43%	-8.58***
	5	0.3993	-7.6656	-9.87%	-4.27***
	22	0.3206	-7.5854	-19.22%	-5.53***

Notes: pooled across all stocks and out-of-sample dates (post 40% warm-up). MSE on log RV^{PK} scale (range-based variance proxy); QLIKE on the variance scale. HLN DM- t is the Harvey–Leybourne–Newbold finite-sample-corrected Diebold–Mariano statistic, computed on the cross-sectional mean loss differential per date with Newey–West HAC variance (bandwidth tied to $\lceil h/5 \rceil - 1$ to handle overlapping multi-step forecasts on a weekly sample stride) and Student- $t(T - 1)$ reference. Positive values indicate the model beats Model A. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Layered ladder: A (HAR core), A_1 (+ VIX, MOVE; HAR-X), A_2 (+ own-stock persistence block), A_3 (+ cross-sectional mean and dispersion of $\hat{d}$), A_4 (+ sector-mean $\hat{d}$), A_5 (+ $\hat{d}$, VIX, MOVE, $\hat{d} \times \text{VIX}$, $\hat{d} \times \text{MOVE}$), C (full union), D (predictors of C estimated by shrinkage / tree-based ML).

The first major result concerns the impact of market-state variables. Adding only the VIX and MOVE controls (Model A_1 , the HAR-X specification) produces substantial forecasting improvements at all horizons: +5.10%, +7.72%, +3.72%. All improvements are statistically significant at conventional levels. This result is economically intuitive. Implied-volatility measures summarize broad market stress conditions and therefore contain substantial forecasting information regarding near-term realized volatility.

The second result is considerably more interesting. Adding only own-stock persistence and roughness variables (Model A_2) produces essentially no forecasting improvement: −0.54%, −0.03%, −0.30%. In several cases the forecasting performance slightly deteriorates relative to the HAR benchmark itself. This finding is important because it clarifies the role of persistence. Own-stock persistence by itself contributes little incremental information beyond the multi-horizon HAR structure. The daily, weekly, and monthly HAR components already capture much of the predictable persistence structure within individual volatility series.

The results change once persistence is aggregated cross-sectionally. Adding only the cross-sectional persistence aggregates (Model A_3) produces small but positive forecasting improvements: −0.10%, +0.37%, +1.26%. Similarly, sectoral persistence aggregates (Model A_4) produce modest positive gains, particularly at the monthly horizon. Although these gains remain relatively small in isolation, their pattern is highly informative. The improvements become larger precisely at longer horizons where uncertainty-duration effects should matter most.

The persistence-by-stress interaction specification (Model A_5) produces substantially larger improvements: +4.91%, +7.63%, +2.87%. At the daily and weekly horizons, these gains closely resemble those obtained by the HAR-X model. At the monthly horizon, however, the interaction structure begins to diverge from the pure market-state specification.

The full structural model (Model C), which combines all persistence blocks simultaneously, produces the strongest overall forecasting performance: +4.55%, +8.24%, +5.54%. The associated HLN-corrected Diebold–Mariano statistics equal approximately +3.15, +3.87, +2.33, implying strong statistical significance at all horizons.

Several aspects of these results deserve emphasis.

First, most of the overall forecasting gain relative to the HAR baseline is captured by adding market-state variables alone through HAR-X. This result is not surprising because implied-volatility indices already summarize substantial information about current market conditions.

Second, persistence features contribute their largest incremental gains at the monthly horizon. Relative to HAR-X, the full structural model adds approximately 1.8 additional percentage points of forecasting improvement at $h = 22$. This pattern is exactly what the structural interpretation predicts. Persistence measures are intended to capture the expected duration of uncertainty propagation rather than merely the level of current volatility. Duration effects should therefore matter more at longer horizons than at very short horizons.

Third, the cumulative loss-differential plots reported in Figure 5 reveal that the forecasting gains accumulate steadily over time rather than arising from a small number of isolated events. The largest improvements occur during the global financial crisis and COVID, precisely when uncertainty-duration effects should become economically most important.

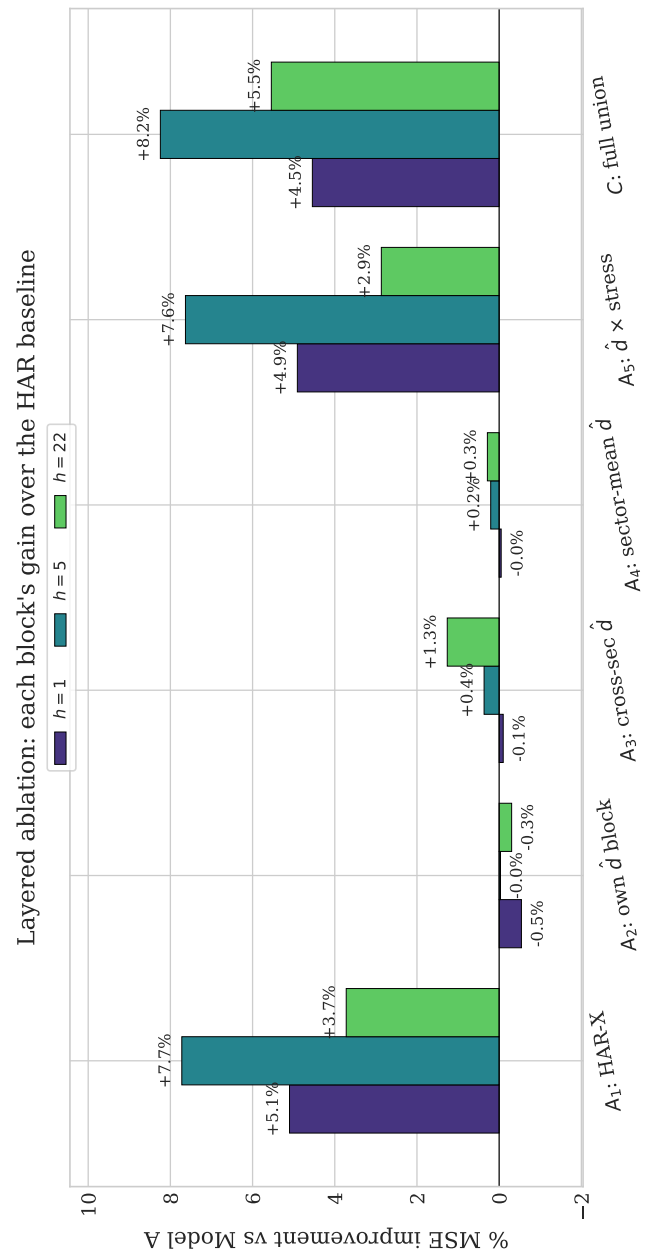


Figure 4 Layered ablation of forecast gains relative to the HAR baseline. Bars report the percentage MSE improvement of each model block over Model A at forecast horizons $h \in \{1, 5, 22\}$.

Cumulative loss differential of Model C against Model A (rising line = C beats A)

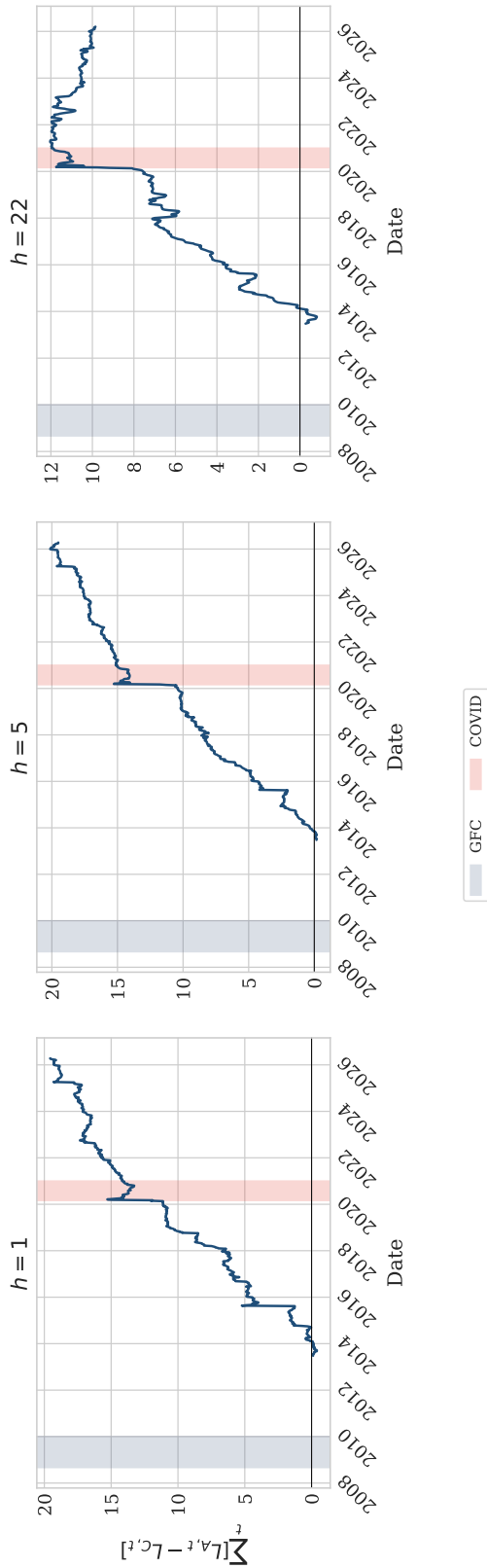


Figure 5 Cumulative loss differential $\sum_t (L_{A,t} - L_{C,t})$ across out-of-sample dates, separately for $h \in \{1, 5, 22\}$. A rising line indicates Model C beats Model A at that date. Crisis bands shade the GFC (2008-Q3 through 2009-Q4) and COVID (March–December 2020).

The comparison between linear and machine-learning estimators is also highly revealing. The shrinkage estimators, Lasso, Ridge, and Elastic Net, track the performance of the full structural model relatively closely at short and medium horizons (+4% to +7%). However, these estimators underperform at the monthly horizon because regularization shrinks the cross-sectional and interaction features more aggressively than appears economically appropriate.

The tree-based estimators perform considerably worse. Random Forest delivers only weak improvements at short horizons and negative performance at the monthly horizon. Gradient Boosting performs especially poorly, producing forecasting deterioration between approximately -10% and -19% . These results carry an important methodological implication. The forecasting gains arise primarily from economically structured feature construction rather than from estimator flexibility itself. The likely explanation is overfitting: the tree-based estimators operate in a relatively small effective sample with approximately 18 predictors and substantial temporal dependence. Under such conditions, flexible nonlinear algorithms may fit noise and local idiosyncrasies rather than stable economic relationships.

Figure 6 further illustrates the breadth of the forecasting gains across the cross-section of stocks. At the weekly horizon, 99% of stocks are improved by HAR-X relative to the HAR benchmark, while approximately 94% are improved by the full structural model. By contrast, Gradient Boosting outperforms HAR for only a very small fraction of the cross-section. This result demonstrates that the persistence-based forecasting gains are not driven by a few isolated securities. Rather, the gains appear broadly throughout the cross-section of equities.

The layered forecasting results therefore support several central conclusions. First, persistence behaves like an economically meaningful state variable rather than merely a statistical nuisance parameter. Second, persistence contributes most strongly during long-horizon and stress-regime forecasting problems, precisely where uncertainty-duration effects should matter most. Third, economically interpretable persistence structures appear more important than highly flexible nonlinear estimation procedures.

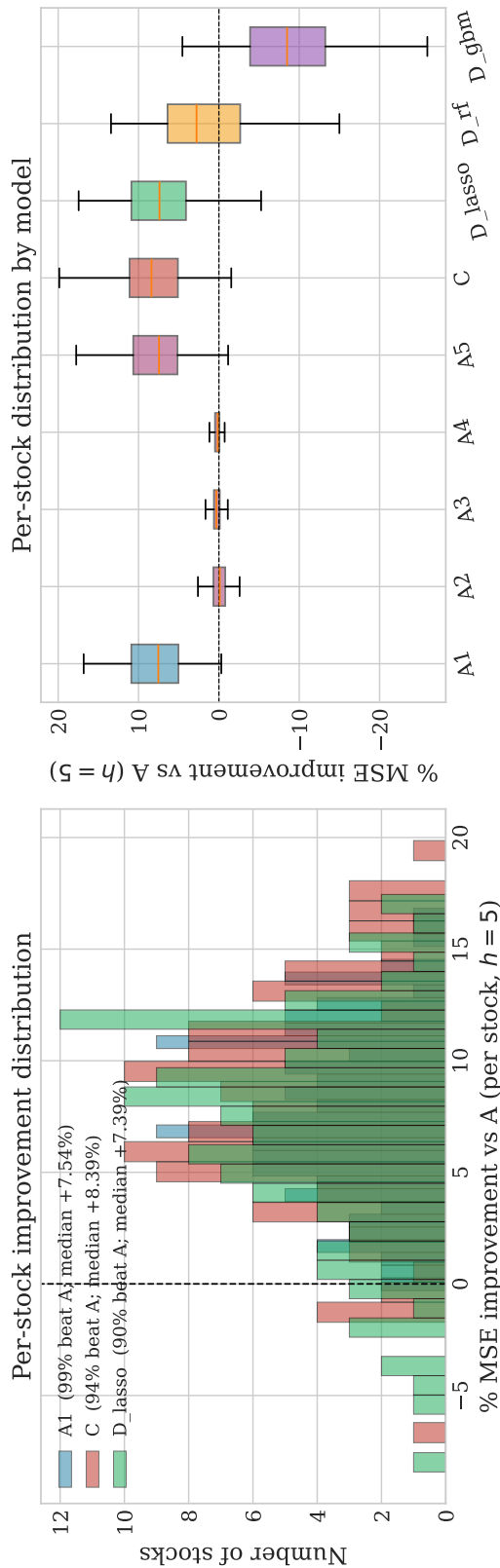
Per-stock forecast-improvement distribution (115 stocks, $h = 5$)

Figure 6 Per-stock MSE improvement vs Model A at $h = 5$. Left panel: histograms for A_1 , C , and D -Lasso. Right panel: boxplot of the per-stock improvement distribution across the full ladder.

Feature Importance

The forecasting results establish that persistence-related variables improve predictive performance, particularly during stress regimes and at longer horizons. The next question concerns which components of the persistence feature vector actually drive these gains. To address this issue, the paper examines standardized Lasso coefficients estimated on the full pooled out-of-sample panel.

Table 7 reports the resulting coefficient estimates separately for the daily, weekly, and monthly forecasting horizons. Because all predictors are standardized prior to estimation, the coefficients are directly comparable in magnitude.

Table 7 Standardised Lasso Coefficients on the Full Pooled Sample

Feature	$h = 1$	$h = 5$	$h = 22$
har_d.log	+0.1330	+0.0991	+0.0675
har_w.log	+0.2106	+0.1840	+0.1460
har_m.log	+0.3540	+0.4047	+0.4499
ret_lag1	-0.0387	-0.0474	-0.0424
ret_lag1_abs	+0.0241	+0.0153	+0.0177
d_gph	+0.0879	+0.0837	+0.0527
delta_d_gph	+0.0113	+0.0069	+0.0029
vol_d_gph	-0.0253	-0.0394	-0.0465
trend_d_gph	-0.0076	-0.0076	-0.0000
h	-0.0169	-0.0091	-0.0060
delta_h	+0.0043	+0.0153	+0.0094
cs_mean_d	-0.0672	-0.0474	-0.0148
cs_std_d	-0.0067	-0.0058	-0.0222
sector_mean_d	+0.0147	+0.0085	+0.0095
vix	+0.2784	+0.2247	+0.1132
move	-0.0315	+0.0000	+0.0214
d_x.vix	-0.1453	-0.1432	-0.1151
d_x.move	+0.0227	+0.0239	+0.0381

Notes: Pooled-Lasso fit on standardised features (full sample, all stocks, all out-of-sample dates), one fit per horizon. Coefficients are on standardised inputs and so are directly comparable in magnitude. Zero coefficients indicate features dropped by Lasso.

Several important patterns emerge immediately.

First, the dominant predictors remain the classical HAR components: $\log RV^{(d)}$, $\log RV^{(w)}$, $\log RV^{(m)}$. Among these variables, the monthly HAR component consistently carries the largest coefficient magnitude, particularly at longer forecasting horizons. This result is economically intuitive because long-horizon realized-volatility information should matter most for medium- and long-horizon volatility forecasting. The persistence of the HAR structure throughout all specifications reinforces an important point emphasized earlier: traditional multi-horizon volatility dynamics remain central even when persistence and roughness variables are introduced. The persistence framework therefore complements rather than replaces conventional HAR-type forecasting structures.

Second, the market-state variables, especially the VIX, also carry a substantial coefficient magnitude. The standardized VIX coefficient remains strongly positive across all horizons, confirming that broad implied-volatility conditions provide important predictive information regarding future realized volatility.

The third and most important finding concerns the persistence aggregates themselves. Cross-sectional persistence measures such as $\bar{d}_t$ and σ_d^t enter consistently across horizons with economically meaningful coefficient magnitudes. Sectoral persistence averages

$\bar{d}_{s(i),t}$ also remain significant contributors to the forecasting system. This result is crucial because it demonstrates that persistence becomes economically informative primarily when viewed as a system-wide or sector-wide state variable rather than as an isolated individual-stock coefficient.

By contrast, the own-stock persistence estimate $\hat{d}_{i,t}$ is not eliminated by the Lasso procedure, but its role should be interpreted cautiously. The layered forecasting results show that own-stock persistence alone contributes little incremental forecasting gain once the HAR components are already included. Thus, the main economic contribution of the persistence block does not come from $\hat{d}_{i,t}$ in isolation, but from the way persistence is combined with market-wide stress, cross-sectional aggregation, sectoral persistence states, and persistence-by-stress interactions.

The persistence-by-stress interaction variable $\hat{d}_t \cdot \text{VIX}_t$ also enters prominently throughout the forecasting system. Its coefficient remains economically large across horizons, confirming that the forecasting impact of market stress depends critically on the persistence state of the financial system. This interaction structure is one of the central empirical findings of the paper. High volatility by itself does not necessarily imply prolonged uncertainty. However, high volatility occurring simultaneously with elevated persistence indicates that uncertainty is likely to dissipate slowly rather than quickly reverting to normal conditions.

The persistence-by-MOVE interaction also contributes positively, although somewhat less strongly than the VIX interaction. This result suggests that persistence interacts not only with equity-market stress but also with broader fixed-income uncertainty conditions.

Several additional patterns deserve mention. The volatility-of-persistence measure $\text{Vol}(\hat{d}_t)$ enters negatively at longer horizons. Economically, this may indicate that highly unstable persistence environments reduce the predictive reliability of persistence itself. The trend component $\text{Trend}(\hat{d}_t)$ generally receives relatively small coefficients, suggesting that the level of persistence matters more than its short-run directional movement. The roughness variables H_t and ΔH_t also contribute more modestly than the persistence aggregates, clarifying the relative roles of roughness and persistence. Roughness captures local short-horizon irregularity, whereas persistence aggregates appear to contain more economically relevant information regarding medium-horizon uncertainty propagation.

Taken together, the feature-importance analysis provides a coherent economic narrative. The dominant predictive structure remains the traditional HAR multi-horizon volatility decomposition. Market-state variables such as VIX provide substantial additional information regarding current stress intensity. Persistence becomes economically important primarily through its aggregate system-wide dimension and through its interaction with market stress conditions. This result strongly supports the structural interpretation developed throughout the paper. Persistence functions as a state-dependent measure of how broadly and how persistently uncertainty is propagating throughout the financial system. The feature importance analysis also helps explain why the full structural model outperforms the more flexible tree-based procedures: the forecasting gains arise from a carefully structured economic feature space rather than from highly nonlinear algorithmic complexity.

Regime Split

The structural interpretation developed earlier implies that persistence measures should be most valuable precisely during periods in which uncertainty resolves slowly. To evaluate this prediction directly, the forecasting results are decomposed across several distinct market regimes: low-volatility states, high-volatility states, and the COVID crisis period.

Table 8 reports the percentage mean squared error improvement of each forecasting model relative to the HAR baseline within these regimes. The results are striking and strongly support the economic interpretation of persistence.

During low-VIX periods, the full structural model produces relatively modest improvements over the HAR benchmark: +2.0%, +6.6%, +9.7% for the daily, weekly, and monthly horizons respectively. By contrast, during high-VIX periods the improvements increase dramatically: +11.8%, +17.9%, +8.8%. The COVID period produces similarly large gains: +6.8%, +18.9%, +17.5%. These differences are economically very large. The forecasting gains during crisis states are often roughly three times larger than those observed during calm volatility regimes.

Equally important is the decomposition of these gains across the layered forecasting ladder. At the weekly horizon during high-VIX states, HAR-X alone already produces substantial gains of +16.1%. However, the full structural model extends this improvement to approximately +17.9%. The difference, approximately 1.9 additional percentage points, represents the incremental contribution of the persistence aggregates and persistence-by-stress interaction terms. A similar pattern emerges during COVID at the monthly horizon. HAR-X produces approximately +15.8% improvement, while the full structural model increases this to approximately +17.5%. Again, persistence contributes meaningful incremental forecasting information beyond standard market-state variables alone.

These results are exactly what the structural interpretation predicts. If the cross-sectional persistence state $\bar{d}_t$ measures the duration of unresolved uncertainty in the financial system, then persistence-related variables should matter most precisely when uncertainty is elevated and slow to dissipate. The gains are not concentrated within a few isolated event windows. Rather, the improvements persist systematically throughout prolonged stress periods, strongly suggesting that the forecasting gains are capturing genuine economic dynamics rather than temporary statistical artifacts.

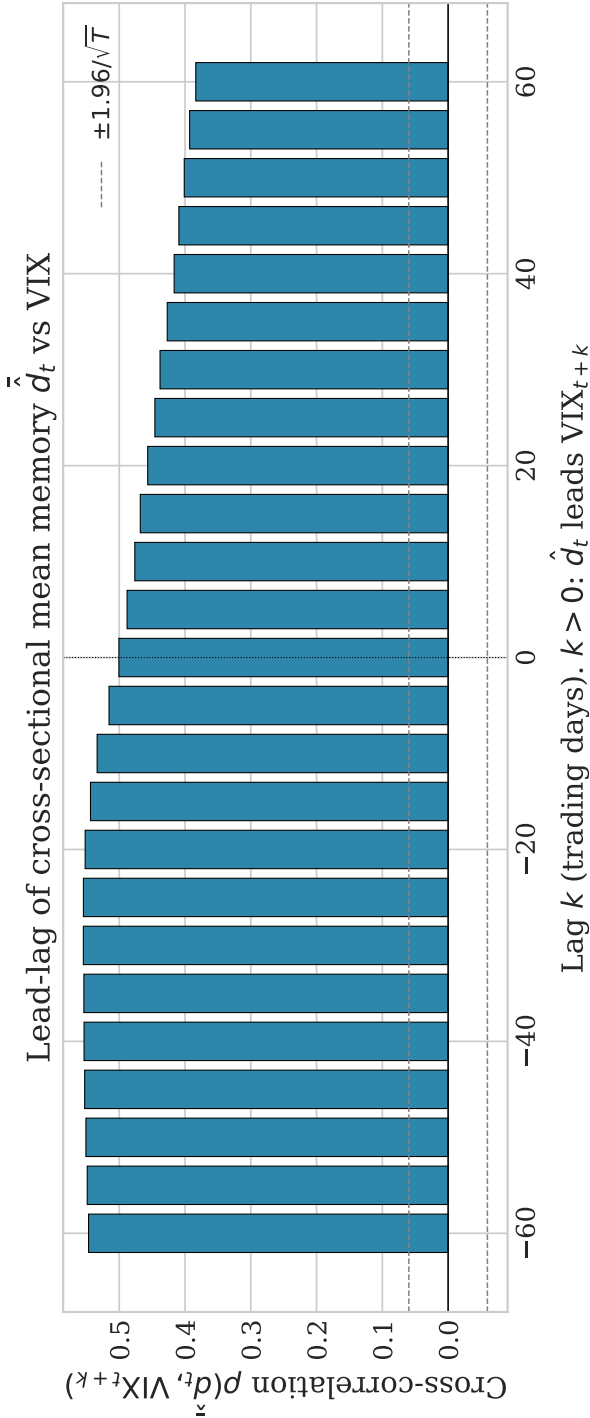
Figure 7 provides an additional diagnostic by examining lead-lag correlations between the cross-sectional persistence state and the VIX. The correlation function peaks near zero lag and remains approximately symmetric around the contemporaneous point. This result indicates that persistence and market stress co-move primarily contemporaneously rather than through strong lead-lag relationships. The absence of strong directional asymmetry suggests that persistence is functioning as a coincident uncertainty-duration indicator rather than a simple leading indicator of market volatility.

The regime analysis therefore provides some of the strongest evidence supporting the economic interpretation of persistence developed earlier in the paper. Persistence-based forecasting variables contribute relatively little during calm markets because uncertainty resolves quickly and the HAR structure already captures most predictable volatility behavior. During prolonged stress episodes,

Table 8 Regime-Conditioned MSE Improvement (%) vs Model A

Regime	h	A	A1	A2	A3	A4	A5	C	D_lasso	D_ridge	D_en	D_rf	D_gbm
COVID (2020)	1	+0.00%	+11.81%	-1.52%	-0.36%	-1.41%	+9.12%	+6.83%	+6.84%	+4.56%	+7.25%	-8.04%	-10.59%
	5	+0.00%	+20.67%	+0.07%	-0.51%	-2.52%	+20.19%	+18.88%	+16.43%	+13.72%	+16.14%	-0.48%	+5.41%
	22	+0.00%	+15.81%	-3.74%	+0.05%	-5.82%	+14.49%	+17.49%	+0.52%	+0.49%	+0.70%	-7.04%	-5.74%
High VIX (Q4)	1	+0.00%	+11.45%	-1.30%	+0.32%	-0.51%	+11.23%	+11.84%	+9.72%	+8.29%	+9.56%	+3.53%	-0.52%
	5	+0.00%	+16.05%	-0.66%	+0.90%	-0.49%	+16.56%	+17.94%	+15.01%	+13.51%	+14.90%	+6.31%	+5.56%
	22	+0.00%	+6.87%	-1.10%	+1.03%	-1.47%	+6.36%	+8.75%	+0.98%	+0.68%	+1.29%	-7.62%	-16.68%
Low VIX (Q1)	1	+0.00%	+3.03%	-0.20%	+0.14%	+0.41%	+3.07%	+2.03%	+1.53%	+1.16%	+1.53%	-2.85%	-21.87%
	5	+0.00%	+5.58%	+0.88%	+1.75%	+1.24%	+5.75%	+6.57%	+5.29%	+4.95%	+5.35%	-1.31%	-17.32%
	22	+0.00%	+5.24%	+0.78%	+2.94%	+1.73%	+5.18%	+9.71%	+3.69%	+3.90%	+3.87%	-6.16%	-17.36%

Notes: Out-of-sample MSE improvement on $\log RV^{PK}$ relative to Model A, computed within each regime. VIX quartiles are computed on the out-of-sample evaluation window. Crisis windows are 2008-Q3 to 2009-Q4 (GFC) and Mar-Dec 2020 (COVID). Positive values indicate the model beats HAR within the regime.



Caveat: $\hat{d}_t$ uses a 750-day backward-looking window, so persistence at small lags reflects window smoothing.

Figure 7 Lead-lag cross-correlation between cross-sectional mean memory $\hat{d}_t$ and the VIX over ± 60 trading days.

however, persistence aggregates become highly informative because they summarize how slowly the financial system is absorbing uncertainty. These findings strongly reinforce the interpretation of persistence as a state-dependent measure of uncertainty duration rather than merely a statistical parameter governing autocorrelation decay.

Sector Split

The regime analysis established that persistence features become most valuable during broad stress episodes. The next question concerns whether the forecasting gains are distributed uniformly across industries or whether certain sectors benefit more strongly from persistence-based information.

Table 9 reports the sector-by-sector forecasting improvements for the weekly forecasting horizon $h = 5$, measured relative to the HAR benchmark. The results reveal substantial heterogeneity across sectors, but every GICS sector exhibits positive gains under the full structural model.

The strongest improvements occur in Materials, Industrials, Real Estate, and Financials. The percentage mean squared error reductions for these sectors equal approximately +12.7%, +10.5%, +10.3%, +9.5%. By contrast, the smallest improvements occur in Information Technology and Energy, with gains approximately equal to +4.2% and +4.9%. Although still positive, these improvements are materially smaller than those observed in the balance-sheet-sensitive sectors.

These sectoral patterns are economically highly plausible. Financials and Real Estate are naturally sensitive to systemic funding conditions, leverage cycles, collateral constraints, and broad credit-market stress. Their volatility dynamics therefore depend heavily on prolonged uncertainty propagation throughout the financial system. Industrials and Materials display similar behavior because they are highly exposed to macroeconomic cycles, global demand conditions, and prolonged investment uncertainty. Volatility shocks in these sectors often propagate slowly through inventories, capital expenditures, commodity markets, and financing conditions.

By contrast, Information Technology volatility is often driven more strongly by firm-specific innovation shocks, product announcements, and idiosyncratic news flow rather than by prolonged systemic stress propagation. Persistence aggregates therefore contribute less incremental forecasting information once conventional volatility predictors are already included. The Energy sector provides another informative case. Energy volatility is heavily influenced by oil-price dynamics and commodity-specific macroeconomic factors that are not explicitly modeled within the forecasting framework. Persistence-based financial-system variables therefore explain a smaller share of volatility behavior in this sector.

Figure 8 extends the sectoral comparison across all three forecasting horizons. First, the sectoral ranking remains remarkably stable across horizons. Materials, Industrials, Real Estate, and Financials remain among the strongest beneficiaries throughout the daily, weekly, and monthly horizons.

Second, the gains generally become larger at the monthly horizon for the balance-sheet-sensitive sectors. For example, Materials displays approximately +12.6% improvement at $h = 22$, while Industrials and Health Care also exhibit particularly strong long-horizon gains. This pattern especially reinforces the structural interpretation: persistence-based variables appear most useful

Table 9 Sector-Level MSE Improvement (%) vs Model A at $h = 5$

Sector	A	A1	A2	A3	A4	A5	C	D_lasso	D_rf	D_gbm
Communication Services	+0.00%	+7.29%	-0.76%	+0.42%	+0.38%	+7.37%	+6.39%	+5.76%	+0.53%	-14.61%
Consumer Discretionary	+0.00%	+8.36%	-0.03%	+0.11%	+0.26%	+7.44%	+8.51%	+6.97%	+1.66%	-9.57%
Consumer Staples	+0.00%	+7.22%	+0.02%	+0.51%	+0.33%	+6.83%	+8.55%	+6.46%	+0.86%	-10.85%
Energy	+0.00%	+4.28%	+0.71%	+0.13%	+0.14%	+5.20%	+4.91%	+1.73%	-2.84%	-13.14%
Financials	+0.00%	+9.45%	+0.27%	+0.19%	+0.19%	+9.46%	+9.48%	+7.79%	+4.24%	-6.76%
Health Care	+0.00%	+6.62%	-0.60%	+0.61%	+0.11%	+7.12%	+8.75%	+7.87%	+3.20%	-9.00%
Industrials	+0.00%	+9.10%	-0.29%	+0.79%	+0.25%	+9.41%	+10.51%	+8.30%	+3.39%	-8.15%
Information Technology	+0.00%	+5.07%	-0.26%	+0.15%	+0.26%	+4.55%	+4.19%	+3.51%	-0.61%	-14.55%
Materials	+0.00%	+10.99%	-0.25%	+0.61%	+0.01%	+10.89%	+12.72%	+11.28%	+4.10%	-12.16%
Real Estate	+0.00%	+10.22%	+0.79%	-0.04%	-0.09%	+10.16%	+10.29%	+10.50%	+5.66%	-7.99%
Utilities	+0.00%	+8.08%	+1.45%	+0.23%	+0.22%	+6.98%	+7.62%	+9.15%	+2.58%	-4.60%

Notes: Out-of-sample MSE improvement on $\log RV^{PK}$ at horizon $h = 5$, by GICS sector, relative to Model A. Pooled across stocks within each sector and over the entire out-of-sample window.

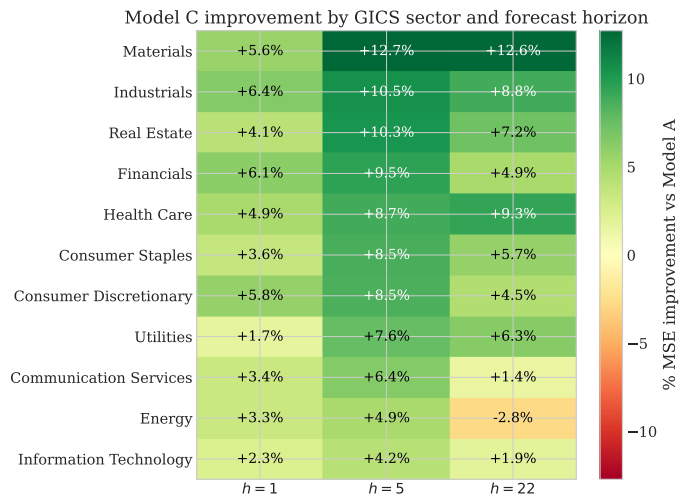


Figure 8 Percentage MSE improvement of Model C over Model A , by GICS sector and forecast horizon. Sectors are sorted by the $h = 5$ gain.

precisely in sectors where uncertainty propagation is likely to persist over extended horizons.

Third, several sectors display asymmetric horizon behavior. Utilities, for example, show relatively modest short-horizon improvements but stronger medium- and long-horizon gains. This may reflect the gradual adjustment dynamics associated with regulated industries and slower-moving macroeconomic factors. The sector analysis also provides additional evidence against purely statistical interpretations of the forecasting gains. If the persistence variables were merely overfitting generic volatility dynamics, one would expect relatively uniform gains across sectors. Instead, the gains align closely with economically plausible sectoral exposure to systemic stress propagation, leverage conditions, and uncertainty duration.

The layered forecasting ladder further clarifies the source of these gains. In nearly every sector, HAR-X alone captures a substantial portion of the overall improvement. However, the full structural model consistently extends the gains beyond HAR-X, especially within the sectors most exposed to prolonged systemic uncertainty. The shrinkage estimators generally track the sectoral ranking reasonably well, although their gains are somewhat attenuated. The tree-based estimators again perform poorly across nearly all sectors, with Gradient Boosting frequently generating strongly negative forecasting performance. These results reinforce one of the central methodological conclusions of the paper: the forecasting gains arise primarily from economically interpretable persistence structures rather than from flexible nonlinear estimation procedures.

Robustness

The final step in the empirical analysis is to determine whether the forecasting improvements remain stable across alternative estimation procedures, rolling-window choices, volatility proxies, liquidity conditions, and benchmark specifications.

Table 10 summarizes a broad robustness battery for the headline comparison between the full structural model and the HAR benchmark at the weekly forecasting horizon. The results are remarkably stable across all perturbations considered.

Table 10 Robustness Summary: Model C vs Model A, $h = 5$

Variant	MSE	% Δ vs A	HLN DM- t
Headline (Model C, full panel)	0.3334	+8.24%	+3.87
Inference: plain DM (illustrative)	0.3334	+8.24%	+3.87 (plain +23.97)
Benchmark: GARCH(1,1) on returns	0.7033	-93.54%	-20.30
Liquidity: low-liquidity half	0.3342	+8.95%	+4.26
Liquidity: high-liquidity half	0.3327	+7.51%	+3.33
Estimator: $\hat{d}_{LW}$ replaces $\hat{d}_{GPH}$	0.3336	+8.21%	+3.85
Rolling window: 500 days	0.3302	+9.13%	+4.00
Rolling window: 1000 days	0.3319	+8.67%	+3.63
Target: log mean future squared returns	0.9102	+4.16%	+3.10
Regime: high VIX (Q_4)	-	+17.94%	-
Regime: low VIX (Q_1)	-	+6.57%	-
Regime: COVID 2020	-	+18.88%	-

Notes: Robustness checks for Model C against Model A at $h = 5$ on $\log RV^{PK}$. Liquidity halves are formed from the static median of mean inverse dollar volume across the sample. $\hat{d}_{LW}$ replaces $\hat{d}_{GPH}$ in the entire feature block (including memory dynamics and interactions). Window variants refit the rolling LRD estimation at the alternative window before regenerating the full feature stack. The target variant replaces $\log RV^{PK}$ with log mean future squared returns (Models A and C both refit). The GARCH(1,1) row fits a constant-mean GARCH(1,1) on log returns with refit every 20 sample steps and reports the resulting log mean conditional variance forecast against the same Parkinson target; its forecast scale differs from the Parkinson RV target (returns variance vs. range-based variance proxy), so the negative $\% \Delta$ vs A reflects both this level mismatch and the well-known limitation of returns-only GARCH for forecasting realised variance. Regime rows come from the Table 7 split at $h = 5$. The plain-DM stat is the unscaled iid pooled-cell statistic; the HLN stat is the panel-aware Harvey-Leybourne-Newbold finite-sample-corrected version.

Replacing the GPH persistence estimator throughout the feature set with the local-Whittle estimator produces forecasting improvements almost identical to the headline specification: +8.21% versus the baseline +8.24%. This result is important because it demonstrates that the forecasting gains are not driven by the particular semiparametric estimator used to construct the persistence variables.

The rolling-window robustness checks produce similarly stable results. Using a shorter 500-day rolling estimation window yields approximately +9.13% forecast improvement, while using a longer 1000-day window yields approximately +8.67%. Both estimates remain highly statistically significant. These findings indicate that the forecasting gains are not artifacts of a particular rolling-window length. Persistence-based forecasting variables remain informative across substantially different temporal smoothing structures.

The volatility-proxy robustness checks are also revealing. Replacing the Parkinson realized-variance target with the logarithm of future mean squared returns, a substantially noisier volatility proxy, reduces the magnitude of the forecasting gains but still produces statistically significant improvement: +4.16%. This result suggests that the persistence-based forecasting gains are not specific to the Parkinson volatility estimator itself. However, the smaller improvement also confirms that cleaner volatility measurement substantially improves the ability to extract economically meaningful persistence information.

Liquidity-based sample splits produce additional evidence supporting the structural interpretation. Dividing the sample into high- and low-liquidity halves yields improvements approximately equal to +8.95% and +7.51%. The somewhat larger gains in the low-liquidity sample are economically plausible because volatility persistence should become more important when markets absorb shocks more slowly and liquidity provision is weaker.

The GARCH(1,1) benchmark row warrants comment. A conventional GARCH(1,1) model estimated directly on returns performs substantially worse than the HAR benchmark when evaluated against the Parkinson realized-volatility target: −93.5% relative performance under mean squared error loss. However, under the proxy-robust QLIKE loss function, which is more robust when volatility proxies contain measurement noise, the relative deterioration becomes much smaller (HLN DM- $t = -2.80$). This distinction is important because it highlights a broader econometric issue. Direct comparisons between returns-based conditional variance models and range-based realized-volatility proxies involve substantial scale mismatch and measurement differences. The paper therefore emphasizes that the forecasting comparison should primarily be interpreted within the realized-volatility forecasting environment rather than as a definitive rejection of all conditional-variance models.

Finally, the robustness analysis again illustrates the importance of correct statistical inference. The naive pooled-cell Diebold–Mariano statistic ignoring cross-sectional dependence equals approximately +23.97, whereas the properly panel-aware HLN-corrected statistic equals approximately +3.87. This difference is enormous and demonstrates how severely significance levels may be inflated when within-date cross-sectional dependence is ignored.

Taken together, the robustness analysis strongly supports the central empirical findings of the paper. The persistence-based forecasting gains remain stable across alternative persistence estimators, rolling-window lengths, volatility proxies, liquidity conditions, and evaluation procedures. The gains are therefore

unlikely to be artifacts of a particular econometric specification or isolated sample feature. Most importantly, the robustness results continue to support the broader structural interpretation developed throughout the paper. Persistence measures behave coherently across market states, forecasting environments, and estimation procedures, suggesting that they capture economically meaningful features of uncertainty propagation rather than merely statistical regularities.

Economic Interpretation and Implications

The empirical results of Section 8 are important not simply because they improve a forecasting statistic, but because they clarify the economic meaning of persistence itself. If persistence-based variables systematically improve volatility forecasts across crisis periods, high-stress states, and balance-sheet-sensitive sectors, then persistence should not be interpreted merely as a technical characteristic of a stochastic process. Instead, it should be understood as an empirical indicator of how slowly financial markets absorb and resolve uncertainty.

The central interpretation developed throughout the paper is therefore economic rather than purely statistical. The rolling persistence state summarizes the expected duration of volatility shocks and uncertainty propagation throughout the financial system. Under this interpretation, persistence measures are not simply filters embedded within econometric specifications. They are state variables describing the temporal organization of market stress.

Volatility-Managed Portfolios

The first direct economic application concerns volatility-managed portfolios. Following Moreira and Muir (2017), the paper constructs volatility-managed equity portfolios that scale exposure inversely with forecast variance. For stock i at date t , the managed five-day return is defined as

$$f_{m,i,t} = \frac{c_{m,i}}{\hat{\sigma}_{m,i,t}^2} r_{i,t \rightarrow t+5}, \quad (18)$$

where $\hat{\sigma}_{m,i,t}^2 = \exp(\hat{y}_{m,i,t})$ denotes the model-based forecast variance and $c_{m,i}$ rescales the managed return to preserve unconditional variance comparability with buy-and-hold strategies.

The resulting portfolio evidence is economically highly informative: see Table 11. Over the full sample, the unmanaged equal-weight buy-and-hold portfolio produces a Sharpe ratio approximately equal to 0.73, while the persistence-augmented managed strategy also produces approximately 0.73. At first glance, this result may appear disappointing. However, the full-sample average conceals the central economic role of persistence. The benefits emerge primarily during stress states rather than during calm market conditions.

During COVID, the unmanaged portfolio Sharpe ratio collapses to approximately 0.65, while the persistence-augmented strategy produces a Sharpe ratio of 1.36. Similarly, the certainty-equivalent return for a mean-variance investor with risk aversion parameter $\gamma = 5$ moves from approximately −3.0% under buy-and-hold to approximately +12.3% under the persistence-augmented strategy. These improvements are economically very large.

The layered forecasting structure again helps isolate the source of the gains. In COVID, for example: $\text{Sharpe}(A) = 0.91$, $\text{Sharpe}(A_1) = 1.06$, $\text{Sharpe}(C) = 1.36$. The persistence aggregates therefore generate an incremental Sharpe improvement of roughly 0.30 even after accounting for the market-state effects captured by HAR-X.

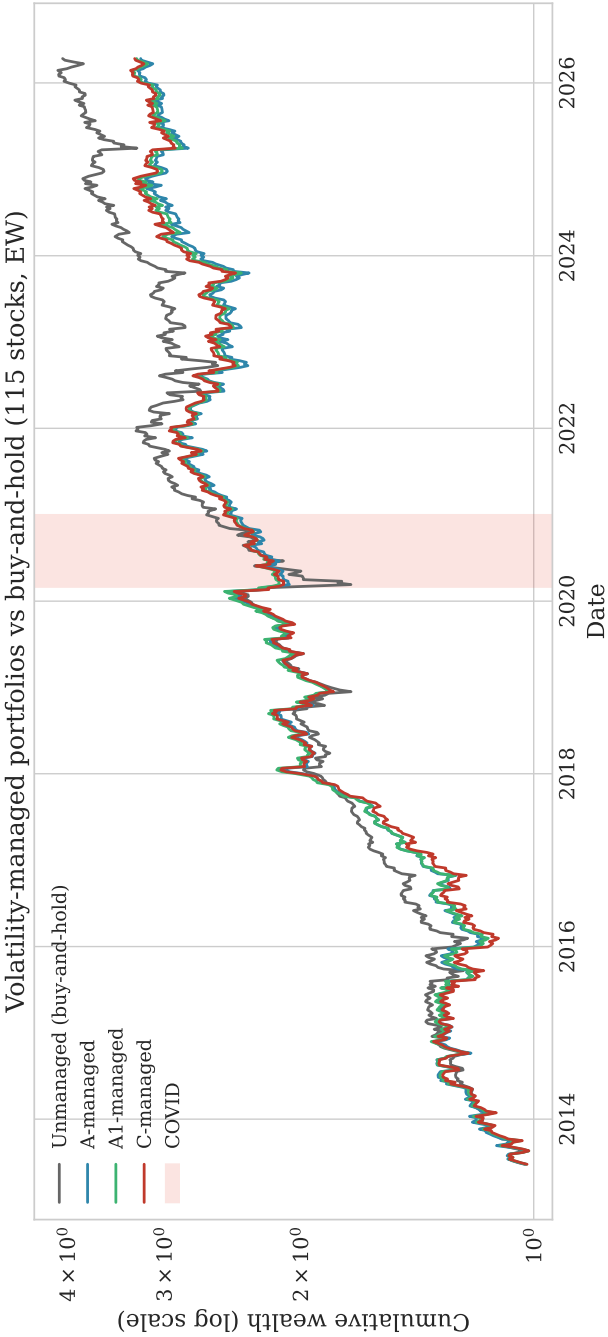


Figure 9 Cumulative wealth (log scale) of the unmanaged equal-weight buy-and-hold portfolio and three volatility-managed variants on the out-of-sample window. Shaded band marks COVID (March–December 2020).

Table 11 Volatility-Managed Portfolios: Risk-Adjusted Performance (Moreira-Muir 2017 Construction)

Regime	Portfolio	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	CER
Full sample	Unmanaged (buy-and-hold)	+10.94%	+14.89%	+0.73	-28.45%	+5.40%
	A-managed	+9.13%	+12.91%	+0.71	-19.89%	+4.96%
	A1-managed	+9.22%	+12.84%	+0.72	-18.56%	+5.10%
	C-managed	+9.27%	+12.72%	+0.73	-17.92%	+5.23%
Low VIX (Q1)	Unmanaged (buy-and-hold)	+6.71%	+7.69%	+0.87	-13.19%	+5.23%
	A-managed	+9.29%	+11.65%	+0.80	-20.54%	+5.90%
	A1-managed	+10.65%	+12.12%	+0.88	-20.18%	+6.98%
	C-managed	+11.37%	+12.41%	+0.92	-20.21%	+7.52%
High VIX (Q4)	Unmanaged (buy-and-hold)	+21.02%	+23.19%	+0.91	-24.12%	+7.58%
	A-managed	+11.63%	+13.55%	+0.86	-16.71%	+7.04%
	A1-managed	+10.87%	+12.39%	+0.88	-14.68%	+7.04%
	C-managed	+11.23%	+12.08%	+0.93	-14.35%	+7.58%
COVID 2020	Unmanaged (buy-and-hold)	+19.62%	+30.07%	+0.65	-11.96%	-2.98%
	A-managed	+11.03%	+12.09%	+0.91	-4.22%	+7.38%
	A1-managed	+11.48%	+10.82%	+1.06	-4.33%	+8.55%
	C-managed	+15.54%	+11.39%	+1.36	-5.04%	+12.30%

Notes: Equal-weight portfolio of 115 stocks, weekly ($h = 5$ trading days) rebalancing aligned to the forecast sample stride. The volatility-managed portfolio uses position weight $w_{m,i,t} = c_{m,i}/\hat{\sigma}_{m,i,t}^2$ where $\hat{\sigma}_{m,i,t}^2 = \exp(\hat{y}_{m,i,t})$ is the model- m forecast of mean RV^{PK} over $[t + 1, t + 5]$. The constant $c_{m,i}$ is set so that $\text{Var}(f_{m,i}) = \text{Var}(r_t)$ at the stock level. Annualized assuming 252/5 five-day periods per year. CER computed for a mean-variance investor with risk aversion $\gamma = 5$.

Figure 9 provides a particularly clear visual interpretation of the mechanism. The managed portfolios sharply reduce exposure precisely during periods in which forecast variance spikes upward. The persistence-augmented strategy cuts exposure most aggressively during the global financial crisis and COVID, preserving capital during the collapse and allowing more efficient participation during the subsequent recovery. This evidence strongly supports the economic interpretation of persistence as a duration-of-stress indicator. Traditional volatility forecasts capture the intensity of contemporaneous risk. Persistence-based variables additionally capture how long elevated uncertainty conditions are likely to remain active.

Persistence as a Temporal Stress Indicator

The empirical evidence suggests that persistence measures summarize a fundamentally different dimension of financial stress than conventional volatility indicators. Two periods may exhibit similar instantaneous volatility levels while differing substantially in the expected duration of uncertainty. One market may experience a brief volatility spike followed by rapid stabilization. Another may experience prolonged uncertainty propagation through funding markets, leverage channels, and institutional balance sheets. Persistence measures appear to distinguish between these environments.

The cross-sectional persistence state $\bar{d}_t$ captures this temporal dimension directly. Empirically, the cross-sectional mean persistence measure rises approximately 68% during the global financial crisis and approximately 86% during COVID relative to the calm 2013–2014 baseline. Its contemporaneous correlation with the VIX equals approximately +0.50. The persistence-by-stress interaction variable strengthens this interpretation even further. For AAPL, for example, the interaction $\hat{d}_t \cdot \text{VIX}_t$ rises from approximately 1.05 during calm periods to approximately 12.16 during the global financial crisis and approximately 8.22 during COVID. These magnitudes are economically enormous and strongly inconsistent with the interpretation of persistence as a purely technical econometric parameter. Instead, persistence appears to summarize how deeply and how persistently uncertainty is propagating throughout the financial system.

Implications for Risk Management and Stress Testing

The forecasting gains at longer horizons carry important implications for risk management. Traditional volatility forecasting systems are central to Value-at-Risk, expected shortfall, margin determination, stress testing, capital allocation, and portfolio scaling. Most such systems focus primarily on the expected magnitude of future volatility rather than on its expected duration. The results of the paper suggest that this may be incomplete.

At the monthly horizon ($h = 22$), the persistence-augmented model improves forecasting performance by approximately +5.5% overall and approximately +17.5% during COVID. These gains indicate that persistence contains forward-looking information about the duration of elevated volatility conditions that is not captured by standard HAR or HAR-X structures. This information is particularly relevant for horizon-sensitive risk-management problems, including:

- multi-day expected shortfall,
- margin calibration,

- stress-duration scenarios,
- liquidity horizon adjustments, and
- multi-period capital allocation.

Persistence therefore provides information not merely about how volatile markets currently are, but about how long elevated risk conditions are likely to persist.

Portfolio Allocation and Volatility Trading

Persistence also has important implications for portfolio management and volatility trading. Volatility-managed strategies implicitly assume that volatility forecasts can distinguish temporary volatility spikes from persistent regime changes. Forecasting systems that fail to distinguish these environments may reduce exposure after temporary shocks and re-enter markets too early, producing costly regime-misclassification errors. Persistence-based variables help mitigate this problem because they summarize the expected duration of uncertainty conditions rather than merely their current intensity.

This interpretation also carries implications for volatility trading across maturities. When contemporaneous volatility is elevated but persistence remains low, uncertainty may dissipate rapidly, favoring shorter-dated volatility exposure. By contrast, a high-persistence state implies that medium-horizon implied volatility may remain elevated for prolonged periods. Persistence therefore provides a potential economic discipline for interpreting volatility-term-structure dynamics.

Macroprudential Monitoring

The persistence framework also suggests broader macroprudential applications. Most macroprudential monitoring systems focus on contemporaneous indicators such as leverage, credit spreads, liquidity conditions, and realized volatility. These indicators measure current stress but do not directly summarize the expected duration of stress propagation. Persistence measures potentially fill this gap.

A simultaneous increase in $\bar{d}_t$ and σ_d^t may indicate that uncertainty is both highly persistent and unevenly distributed throughout the financial system. This combination is especially dangerous because it suggests broad systemic vulnerability together with localized pockets of acute fragility. Although the present paper does not explicitly develop a macroprudential framework, the persistence-state variables introduced here naturally extend to broader financial-stability applications.

Implications for Machine Learning in Finance

Finally, the forecasting comparisons carry broader methodological implications for machine learning in finance. The linear, economically structured forecasting systems consistently outperform the highly flexible tree-based estimators throughout the empirical analysis. Random Forest contributes little incremental value, while Gradient Boosting substantially worsens forecasting performance across nearly all specifications and regimes.

This result strongly suggests that the economic value arises primarily from feature-space construction rather than estimator complexity itself. The persistence variables become useful because they summarize economically interpretable mechanisms, uncertainty duration, stress propagation, cross-sectional fragility, and persistence-by-stress interactions, rather than because nonlinear algorithms discover arbitrary predictive patterns in large feature

spaces. The results therefore support a broader methodological perspective increasingly emphasized in empirical finance: machine learning contributes most effectively when guided by economically interpretable feature engineering rather than by unrestricted algorithmic flexibility alone.

Conclusion

This paper has argued that persistence in financial volatility should not be treated merely as a technical property of a stochastic process. Instead, persistence should be interpreted as an economically meaningful feature of financial markets reflecting how slowly uncertainty is absorbed, how long the effects of shocks remain active, and how market conditions evolve across calm and stressed regimes.

The classical econometric literature introduced persistence through fractional integration, long-memory dependence, and conditional-volatility models, providing rigorous tools for describing slow decay in volatility dynamics. The broader contribution of the present paper has been to reinterpret these measurements from a reduced-form perspective. The memory parameter is not only a coefficient governing autocorrelation decay. It is also a reduced-form indicator of information persistence, liquidity-mediated propagation, heterogeneous adjustment across investors and sectors, and the temporal depth of market stress.

A second central theme has been that volatility dynamics contain multiple temporal layers. At short horizons, volatility exhibits highly irregular local behavior emphasized by the rough-volatility literature (Gatheral et al., 2018). At longer horizons, volatility displays persistent dependence captured by long-memory and multi-horizon forecasting structures such as HAR. These dimensions are not mutually exclusive. The empirical analysis demonstrates that financial volatility is simultaneously rough at short horizons and persistent over longer horizons.

The empirical evidence is remarkably strong. The 115-stock panel covering November 2001 through April 2026 produces cross-sectional GPH and local-Whittle persistence estimates of approximately

$$\hat{d} = 0.226 \quad \text{and} \quad \hat{d} = 0.440,$$

with statistical significance in nearly every stock. At the same time, the rolling Hurst exponent satisfies $H < \frac{1}{2}$ throughout the panel, with a cross-sectional mean near $H = 0.063$. These findings confirm that volatility exhibits both long-horizon persistence and substantial local roughness simultaneously.

A third contribution has been operational and forecasting-oriented. Embedding persistence-based variables into a layered forecasting framework produces a clean empirical attribution of predictive gains. Own-stock persistence by itself contributes little incremental forecasting information because the HAR structure already captures much of the predictable individual volatility persistence. However, cross-sectional persistence aggregates, sectoral persistence states, and persistence-by-stress interactions provide substantial incremental forecasting improvements beyond both HAR and HAR-X benchmarks.

The forecasting improvements are economically and statistically meaningful: 4.6–8.2% out-of-sample MSE reduction across horizons relative to HAR, with significance at conventional levels under panel-aware HLN-corrected Diebold–Mariano inference. The gains become

particularly strong during stress regimes: +17.9% during high-VIX states and +18.9% during COVID at the weekly horizon.

The economic significance extends beyond forecasting statistics themselves. In the Moreira–Muir volatility-managed portfolio exercise, the persistence-augmented forecasting system produces a COVID Sharpe ratio of 1.36 compared with 0.65 for buy-and-hold and 1.06 for the HAR-X-managed benchmark. The persistence aggregates therefore generate a substantial incremental Sharpe improvement during stress states.

Equally important is the methodological implication concerning machine learning in finance. The tree-based machine-learning procedures fail to match the performance of the economically structured forecasting systems. Random Forest contributes little incremental value, while Gradient Boosting substantially worsens forecasting performance. The forecasting gains therefore arise primarily from economically interpretable feature construction rather than from unrestricted nonlinear estimator flexibility. This finding supports a broader methodological lesson increasingly emphasized in empirical finance: machine learning contributes most effectively when guided by economically interpretable feature engineering rather than by purely algorithmic complexity.

The paper also suggests several directions for future research. One natural extension is to enrich the rough-volatility component using longer high-frequency datasets in order to study more carefully how local irregularity interacts with long-horizon persistence across market regimes. Another extension would apply the framework beyond U.S. equities to fixed income, foreign exchange, commodities, and volatility derivatives, where the structural interpretation of persistence may differ but is likely to remain economically informative.

A third direction would integrate the cross-sectional persistence state into multivariate covariance and correlation forecasting systems. Because many practical portfolio-allocation and risk-management problems involve portfolios rather than individual assets, this extension would connect naturally to the macroprudential literature on systemic fragility and financial stability.

More broadly, the results suggest that persistence measures deserve interpretation not merely as statistical objects but as empirical indicators of the temporal organization of financial uncertainty. Markets differ not only in how volatile they are, but also in how long uncertainty remains active once shocks occur. Persistence-based state variables appear capable of measuring this dimension directly.

Conflicts of interest

The authors declare that they have no competing interests.

Funding

This work recieved no funding.

Data availability

The data underlying this article are available through Bloomberg Terminal service.

Author contributions statement

A.D: software and data analysis, methodological conceptualization, writing and editing of manuscript. N.A: Data collection ,methodological conceptualization and structuring, writing, editing, and reviewing. S.T.R: Conceptualization, editing and reviewing, overall supervision.

Acknowledgments

Thank Texas Tech University for their support.

References

- Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625.
- Baillie, R. T., Bollerslev, T., and Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1):3–30.
- Barndorff-Nielsen, O. E. and Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2):253–280.
- Bennedsen, M., Lunde, A., and Pakkanen, M. S. (2021). Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5):961–1006.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics*, 13(3):253–263.
- Ding, Z., Granger, C. W. J., and Engle, R. F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1):83–106.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232.
- Gatheral, J., Jaisson, T., and Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6):933–949.
- Geweke, J. and Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4):221–238.
- Granger, C. W. J. and Joyeux, R. (1980). An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1):15–29.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273.
- Harvey, D., Leybourne, S., and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291.
- Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2 edition.
- Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1):165–176.
- Mikosch, T. and Stărică, C. (2004). Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. *Review of Economics and Statistics*, 86(1):378–390.
- Moreira, A. and Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4):1611–1644.
- Müller, U. A., Dacorogna, M. M., Davé, R. D., Olsen, R. B., Pictet, O. V., and von Weizsäcker, J. E. (1997). Volatilities of different time resolutions—analyzing the dynamics of market components. *Journal of Empirical Finance*, 4(2–3):213–239.
- Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256.
- Robinson, P. M. (1995). Gaussian semiparametric estimation of long range dependence. *Annals of Statistics*, 23(5):1630–1661.